

DATA ANALYSIS PORTFOLIO

PREPARED BY:-
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PROFESSIONAL BACKGROUND

Results-driven Data Analyst with a strong foundation in finance and analytics, backed by an MBA from ICFAI Business School, Kolkata. A former Deputy Manager at ICICI Bank, specializing in wholesale banking for large clients, with 1.5 years of experience in financial analysis and client management. Passionate about data-driven decision-making, with hands-on expertise in SQL Workbench and Microsoft Excel.

Successfully completed 8 data analytics projects, including operational analytics at Microsoft, hiring process data analysis for Trainity, and an IMDB movies analysis. Proficient in data cleaning, exploratory data analysis (EDA), metric investigation, and deriving actionable insights.

In addition to analytics and finance, I am also a digital entrepreneur and network marketer with Forever Living Products, leveraging digital marketing strategies and networking skills to grow businesses online. Experienced in fundraising and volunteering for NGOs, demonstrating strong leadership, communication, and teamwork abilities.

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DATA ANALYSIS STEP IN AUCTION

PROJECT-1

DESCRIPTION:

SELECTING A PLAYER IN AUCTION WE CAN USE DATA ANALYTICS

PROBLEM:

STEPS FOLLOWED IN DATA ANALYTICS

FINDINGS:

STEP-1: PLAN- We First decide which players I have to buy whether it will be a Batsman, Bowler or a wicket keeper

STEP-2: PREPARE- We need to check the amount of purse we have in the bag and the base price of particular players

STEP-3: PROCESS- Here we have to decide if we are going for any particular category of players for an example, we want bowler then we have to look whether we will go for Spin bowler or Fast Bowler then it could be Indian or Foreign players and their availability throughout the season

STEP-4: ANALYZE- Here we will focus on that particular player according to his past 2-3 months performance, then whether he will fit in the team and playing eleven and their availability throughout the season

STEP-5: SHARE- Now we discuss all these things in our think tank and back-end team and find the best for the team

STEP-6: ACT- Finally, we buy him or them



INSTAGRAM USER ANALYTICS

PROJECT 2

Project Description:

- Business insights are found that can be used by teams in the business to launch a new marketing campaign, discussing the features to build an application and track

Creating Database:

- ☐ Values are created and inserted in the database using DDL and DML .
- ☐ SQL queries provided as per project in MYSQL database using MY SQL workbench

FINDINGS:

MARKETING ANALYSIS:

Rewarding Most Loyal Users: People who have been using the platform for the longest time (TOP 5)

```
select username, created_at from users order by created_at limit 5;
```

	username	created_at
▶	Darby_Herzog	2016-05-06 00:14:21
	Emilio_Bernier52	2016-05-06 13:04:30
	Elenor88	2016-05-08 01:30:41
	Nicole71	2016-05-09 17:30:22
	Jordyn.Jacobson2	2016-05-14 07:56:26

Remind inactive users to start posting: By sending them promotional emails to post their First photo. The users who have never posted a single photo on Instagram

```
select * from users u left join photos p on p.user_id=u.id where p.image_url is null order by u.username;
```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 							
id	username	created_at	id	image_url	user_id	created_dat	
5	Aniya_Hackett	2016-12-07 01:04:39	NULL	NULL	NULL	NULL	
83	Bartholome.Bernhard	2016-11-06 02:31:23	NULL	NULL	NULL	NULL	
91	Bethany20	2016-06-03 23:31:53	NULL	NULL	NULL	NULL	
80	Darby_Herzog	2016-05-06 00:14:21	NULL	NULL	NULL	NULL	
45	David.Osinski47	2017-02-05 21:23:37	NULL	NULL	NULL	NULL	
54	Duane60	2016-12-21 04:43:38	NULL	NULL	NULL	NULL	
90	Esmeralda.Mraz57	2017-03-03 11:52:27	NULL	NULL	NULL	NULL	
81	Esther.Zulauf61	2017-01-14 17:02:34	NULL	NULL	NULL	NULL	
68	Franco_Keebler64	2016-11-13 20:09:27	NULL	NULL	NULL	NULL	
74	Hulda.Macejkovic	2017-01-25 17:17:28	NULL	NULL	NULL	NULL	
14	Jadyn81	2017-02-06 23:29:16	NULL	NULL	NULL	NULL	
76	Janelle.Nikolaus81	2016-07-21 09:26:09	NULL	NULL	NULL	NULL	
89	Jessyca_West	2016-09-14 23:47:05	NULL	NULL	NULL	NULL	

	id	username	created_at	id	image_url	user_id	created_dat
	57	Julien_Schmidt	2017-02-02 23:12:48	NULL	NULL	NULL	NULL
	7	Kasandra_Homenick	2016-12-12 06:50:08	NULL	NULL	NULL	NULL
	75	Leslie67	2016-09-21 05:14:01	NULL	NULL	NULL	NULL
	53	Linnea59	2017-02-07 07:49:34	NULL	NULL	NULL	NULL
	24	Maxwell.Halvorson	2017-04-18 02:32:44	NULL	NULL	NULL	NULL
	41	Mckenna17	2016-07-17 17:25:45	NULL	NULL	NULL	NULL
	66	Mike.Auer39	2016-07-01 17:36:15	NULL	NULL	NULL	NULL
	49	Morgan.Kassulke	2016-10-30 12:42:31	NULL	NULL	NULL	NULL
	71	Nia_Haag	2016-05-14 15:38:50	NULL	NULL	NULL	NULL
	36	Ollie_Ledner37	2016-08-04 15:42:20	NULL	NULL	NULL	NULL
	34	Pearl7	2016-07-08 21:42:01	NULL	NULL	NULL	NULL
	21	Rocio33	2017-01-23 11:51:15	NULL	NULL	NULL	NULL
	25	Tierra.Trantow	2016-10-03 12:49:21	NULL	NULL	NULL	NULL

Declaring Context winner:

The team started a contest and the user who gets the most likes on a single photo will win the contest now they wish to declare the winner, Now I have to identify the winner of the contest and provide the details to the team

```
select likes.photo_id,users.username,count(likes.user_id) as nooflikes
from likes inner join photos on likes.photo_id=photos.id
inner join users on photos.user_id=users.id group by
likes.photo_id,users.username order by nooflikes desc;
```

	photo_id	username	nooflikes
▶	145	Zack Kemmer93	48

Hashtag Research: This is to find out which hashtags have to be used to reach the most people on the platform. Now have to identify Top 5 most commonly used hashtags on the platform

```
select t.tag_name, count(p.photo_id) as ht from photo_tags p
inner join tags t on t.id=p.tag_id
group by t.tag_name order by ht desc limit 5;
```

The top 5 most commonly used hashtags on the platform

	tag_name	ht
	smile	59
	beach	42
	party	39
	fun	38
	concert	24

Launch Ad Campaign:

The team wants to know on which day it will be most suitable to launch the Advertisement campaigns. Now to find which day of the week do most users register on to schedule an AD campaign

```
select DATE_FORMAT((created_at), '%w') as dayy,count(username) from users group by 1 order by 2 desc;
```

dayy	count(username)
4	16
0	16
5	15
2	14
1	14
3	13
6	12



User Engagement:

This is to know whether the users are still active and post on Instagram or they are making fewer number of posts

```
1 with base as(select u.id as userid,count(p.id) as photoid  
2 from users u left join photos p on p.user_id=u.id group by u.id)  
3 select sum(photoid) as totalphotos,count(userid) as toal_users,sum(photoid)/count(userid) as photopersuer from base;
```

	totalphotos	toal_users	photopersuer
▶	257	100	2.5700

Bots and Fake Accounts:

The investor wants to know whether the platform is full with fake and dummy accounts

with base as (select u.username, count(l.photo_id) as likess from likes l inner join users u on u.id=l.user_id group by u.username)

select username, likess from base where likess=(select count(*) from photos) order by username;

	username	likess		username	likess
	Aniya_Hackett	257		Julien_Schmidt	257
	Bethany20	257		Leslie67	257
	Duane60	257		Maxwell.Halvorson	257
	Jadyn81	257		Mckenna 17	257
	Janelle.Nikolaus81	257		Mike.Auer39	257
				Nia_Haag	257
				Ollie_Ledner37	257
				Rocio33	257

RESULTS:

- ❖ This gives the fundamentals of data analysis using SQL Queries to extract insights from the database to track how user engage and interact with our platform so that we give proper and insights to the Product, Marketing and Development team

CONCLUSIONS FROM THE ANALYSIS:

- User engagement can be very useful matrix for growth of the company
- Marketing team can reward the most loyal customers, send promotional email to the inactive customers, use popular hashtags and find out the best day to do the ad campaign
- Company can remove the fake and dummy accounts from the platform to increase user experience

OPERATIONAL AND METRIC ANALYTICS

PROJECT 3

PROJECT DESCRIPTION:

The objective of the project is to analyse the company's end to end operations which helps identifying the area for improvement. As the Data Analyst of Microsoft, I will work closely with the marketing, operation and support team to derive valuable insights from the collected data source. The project also focuses on investigating metric spikes in daily engagement and sales, investigating their causes and providing insights to address any dip in these metrics

APPROACH:

1. Project involves understanding and gathering data sets and tables from various sources and departments within company
2. Making sure that the given variable and attribute can be understood properly
3. Importing files into my SQL Workbench and through SQL the collected data will be analysed to give insights and patterns
4. Key performance metrics are such as daily engagement, sales and customer satisfaction will be identified in collaboration with cross-functional teams
5. Cross checking the queries one or two times so that the code will run without errors
6. Once the questions are answered it will be saved with screenshot
7. Finally, it will be added to the document

INSIGHTS:

CASE STUDY 1: JOB DATA ANALYSIS

Jobs Reviewed Over Time:

Task: Write an SQL query to calculate the number of jobs reviewed per hour for each day in November 2020.

```
select ds as date,
count(job_id) as joint_job_id,
round((sum(time_spent) / 3600), 2) as total_time_spent_hr,
round((count(job_id)/ (sum(time_spent) / 3600)), 2) as "jobs reviewd per hour per day" from job_data
WHERE ds BETWEEN "2020-11-01" AND "2020-11-30" group by ds
order by ds;
```

Result Grid

 Filter Rows:

Export: 

Wrap Cell Content: 

	date	joint_job_id	total_time_spent_hr	jobs reviewd per hour per day
►	2020-11-25	3	0.04	80.00
	2020-11-26	3	0.05	64.29
	2020-11-27	3	0.09	34.62
	2020-11-28	6	0.03	218.18
	2020-11-29	3	0.02	180.00
	2020-11-30	6	0.03	180.00

Throughput Analysis:

Task: Write an SQL query to calculate the 7-day rolling average of throughput. Additionally, explain whether you prefer using the daily metric or the 7-day rolling average for throughput, and why.

Weekly Throughput:

```
select round((count(event)/sum(time_spent)),2) as weekly_throughput  
from job_data;
```

	weekly_throughput
▶	0.03

Daily Throughput:

```
# Daily metric throughput  
select ds as date, round((count(event)/sum(time_spent)),2) as daily_metric  
from job_data group by date;
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	date	daily_metric
▶	2020-11-30	0.05
	2020-11-29	0.05
	2020-11-28	0.06
	2020-11-27	0.01
	2020-11-26	0.02
	2020-11-25	0.02

LANGUAGE SHARE ANALYSIS:

Task: Write an SQL query to calculate the percentage share of each language over the last 30 days.

```
• select language, round(100 * count(*) / total ,2) as percentage
  from job_data
  cross join
  (select count(*) as total
   from job_data) as jd
  group by language,jd.total;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	language	percentage
▶	English	12.50
	Arabic	12.50
	Persian	37.50
	Hindi	12.50
	French	12.50
	Italian	12.50

INVESTIGATING METRIC SPIKE

WEEKLY USER ENGAGEMENT:

Task: Write an SQL query to calculate the weekly user engagement.

```
select extract(week from occurred_at) as week_number,  
count(distinct user_id) as active_user  
from events  
where event_type='engagement'  
group by week_number  
order by week_number;
```

	week_number	active_user
►	17	663
	18	1068
	19	1113
	20	1154
	21	1121
	22	1186
	23	1232
	24	1275
	25	1264
	26	1302

	week_number	active_user
	27	1372
	28	1365
	29	1376
	30	1467
	31	1299
	32	1225
	33	1225
	34	1204
	35	104

USER GROWTH ANALYSIS:

Task: Write an SQL query to calculate the user growth for the product.

```
with weekly_active_users as (  
  select  
    extract(year from created_at) as year,  
    extract(week from created_at) as week_number,  
    count(distinct user_id) as num_of_users  
  from users  
  group by year, week_number)
```

```
select  
  year,  
  week_number,  
  num_of_users,  
  sum(num_of_users) over (order by year, week_number) as cumulative_users  
from weekly_active_users  
order by year, week_number;
```

Result Grid Filter Rows: Export: Wrap Cell Content:				
	year	week_number	num_of_users	cumulative_users
	2014	31	193	8598
	2014	32	245	8843
	2014	33	261	9104
	2014	34	259	9363
	2014	35	18	9381

EMAIL ENGAGEMENT ANALYSIS:

Task: Write an SQL query to calculate the email engagement metrics.

```
select
100.0* sum(case when email_action = 'email_open' then 1 else 0 end) /
sum(case when email_action = 'email_sent' then 1 else 0 end) as email_open_rate,
100.0* sum(case when email_action = 'email_clicked' then 1 else 0 end) /
sum(case when email_action = 'email_sent' then 1 else 0 end) as email_clicked_rate
from
(select *,
case
when action in ('sent_weekly_digest', 'sent_reengagement_email') then 'email_sent'
when action in ('email_open') then 'email_open'
when action in ('email_clickthrough') then 'email_clicked'
else null
end as email_action
from
project3.email_events
) a;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	email_open_rate	email_clicked_rate
▶	33.58339	14.78989

HIRING PROCESS ANALYTICS

PROJECT 4

PROJECT DESCRIPTION:

The project was designed to conduct data analytics on the hiring process within a company, with the primary objective of gaining insights into various aspects such as gender distribution, salary analysis, departmental composition, and position tiers. By analyzing these factors with the desired visualization charts the project aimed to provide valuable insights into hiring patterns

APPROACH:

To perform the project a systematic approach was followed. A dataset having relevant information on hires, including gender, salary, department, and position, was obtained. Microsoft Excel 2021 was selected as the primary tool for data analysis due to its versatility and robust capabilities in handling tabular data. Specific techniques such as pivot tables, charts, and formulas were employed to analyze the dataset and extract meaningful insights.

TECH-STACK USED:

Software: Microsoft Excel 2021

Purpose: Excel was chosen for its extensive data analysis functions, including pivot tables, charts, and statistical functions and visualization of the hiring data

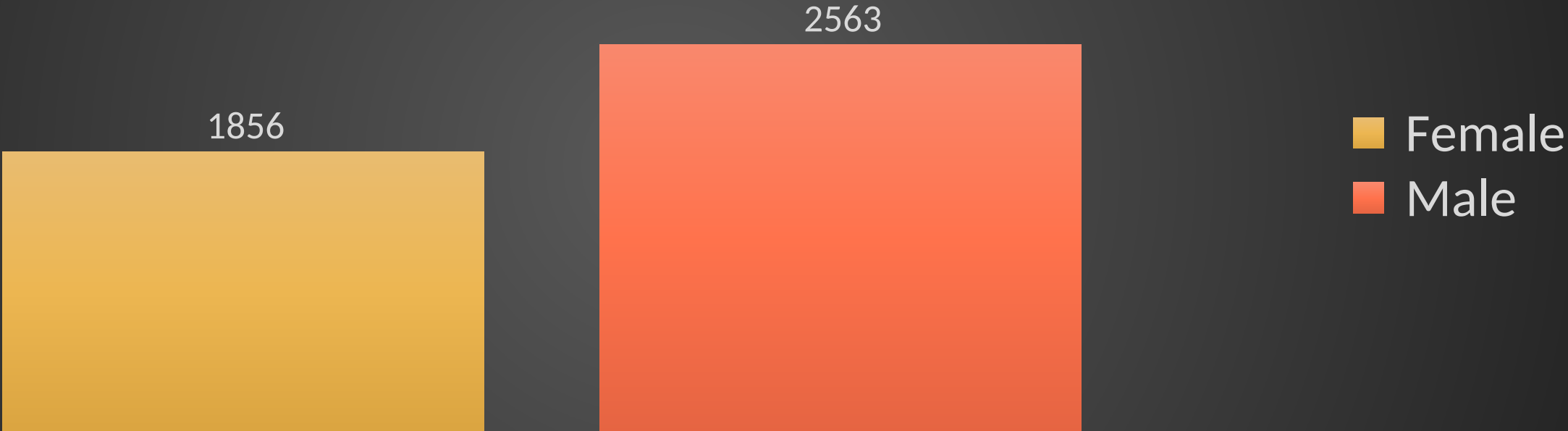
TASKS:

A. Hiring Analysis: The hiring process involves bringing new individuals into the organization for various roles.

Your Task: Determine the gender distribution of hires. How many males and females have been hired by the company?

Count of event_name	event_name		
Status	Female	Male	Grand Total
Hired	1856	2563	4419
Grand Total	1856	2563	4419

HIRED MALE AND FEMALE

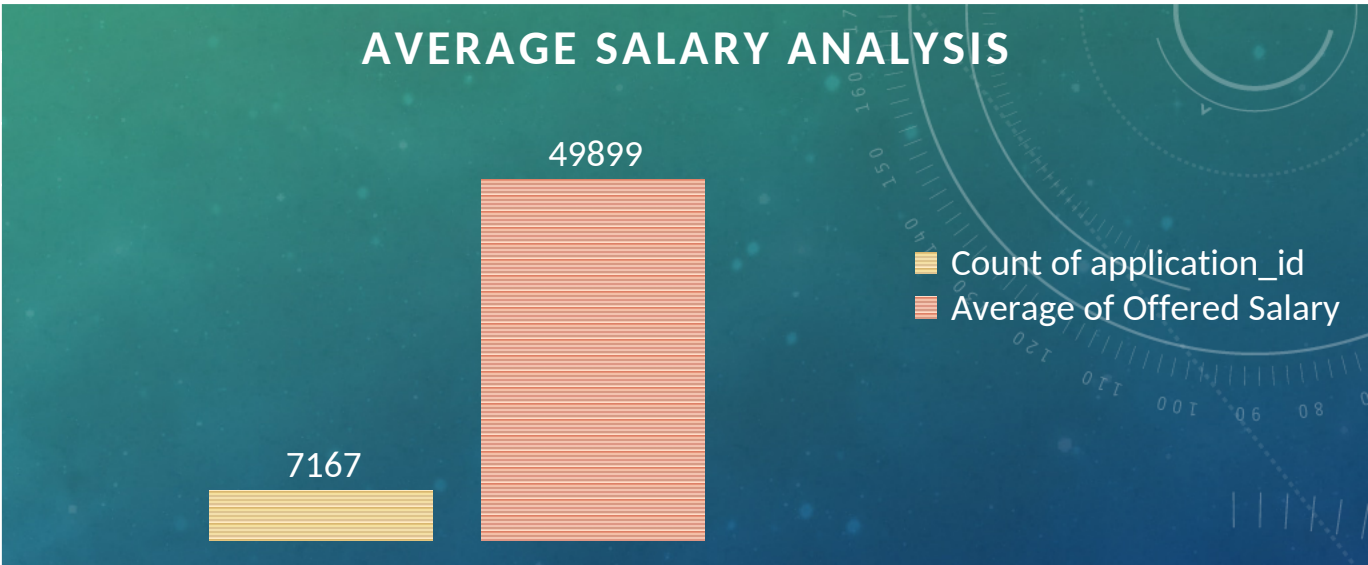


B. Salary Analysis:

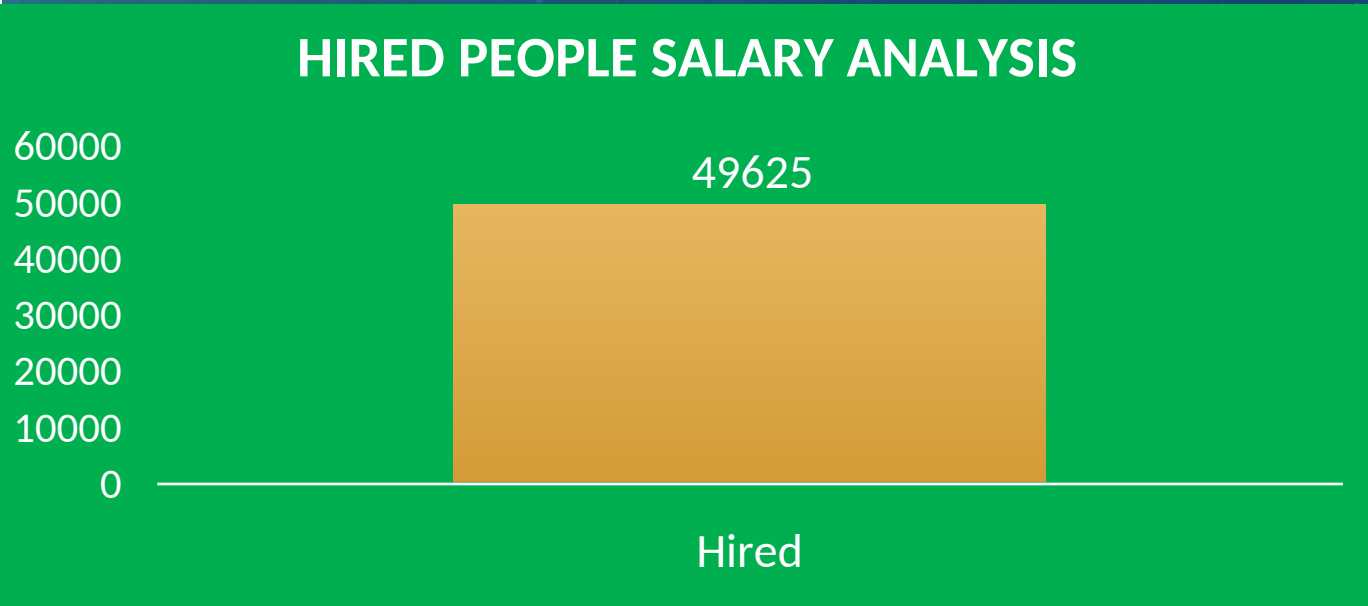
The average salary is calculated by adding up the salaries of a group of employees and then dividing the total by the number of employees.

Your Task: What is the average salary offered by this company? Use Excel functions to calculate this.

BY PIVOT TABLE	
Count of application_id	Average of Offered Salary
7167	49899



Status	Average of Offered Salary
Hired	49625
Grand Total	49625

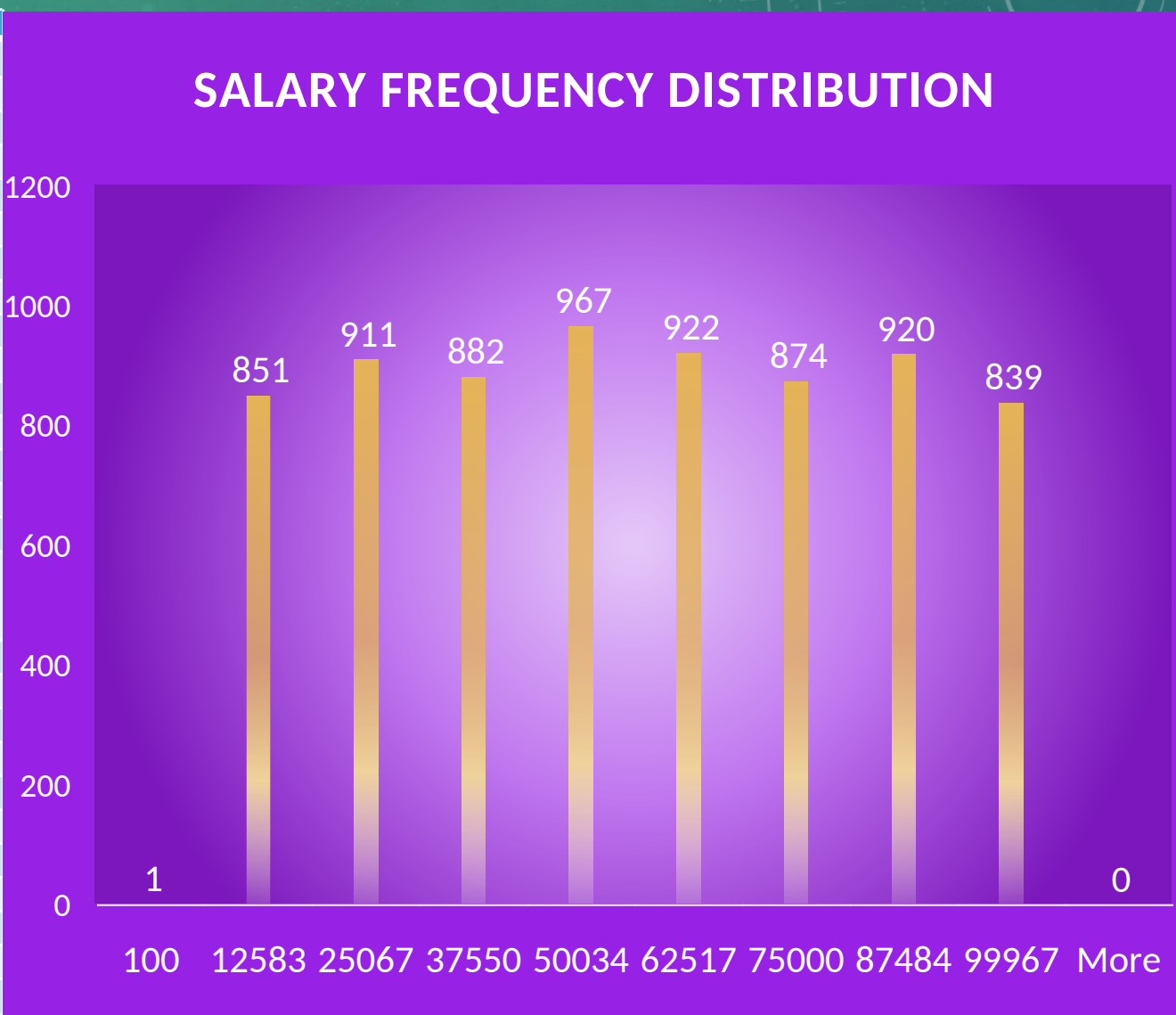


C. Salary Distribution:

Class intervals represent ranges of values, in this case, salary ranges. The class interval is the difference between the upper and lower limits of a class.

Your Task: Create class intervals for the salaries in the company. This will help you understand the salary distribution

Column1	
Mean	49899.3
Standard Error	335.0679
Median	49625
Mode	99967
Standard Deviation	28366.22
Sample Variance	8.05E+08
Kurtosis	-1.17945
Skewness	0.013242
Range	99867
Minimum	100
Maximum	99967
Sum	3.58E+08
Count	7167
99867	RANGE
Bin	Frequency
100	1
12583	851
25067	911
37550	882
50034	967
62517	922
75000	874
87484	920
99967	839
More	0

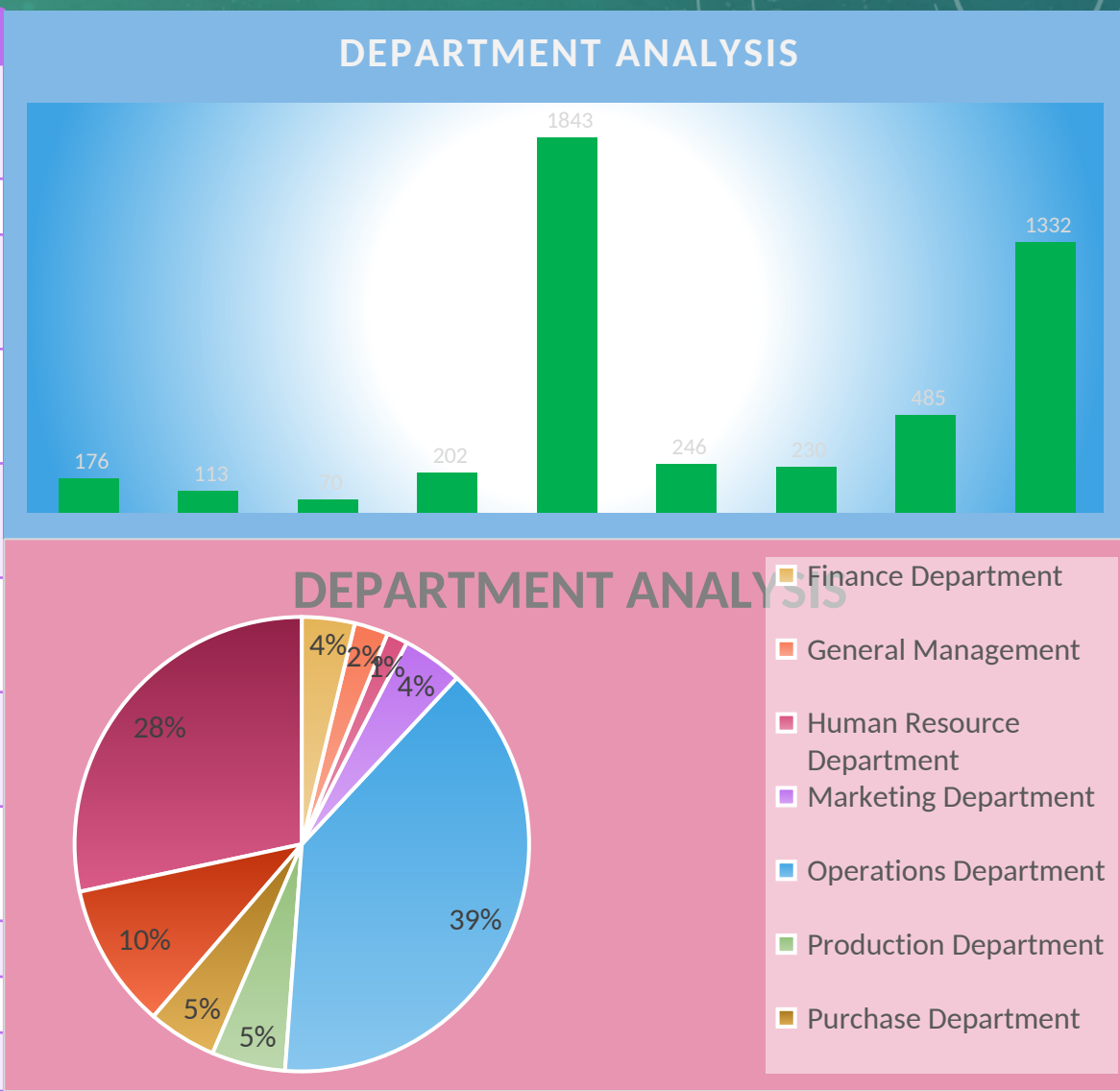


D. Departmental Analysis:

Visualizing data through charts and plots is a crucial part of data analysis.

Your Task: Use a pie chart, bar graph, or any other suitable visualization to show the proportion of people working in different departments.

Count of Status		Status
Department	Hired	Grand Total
Finance Department	176	176
General Management	113	113
Human Resource Department	70	70
Marketing Department	202	202
Operations Department	1843	1843
Production Department	246	246
Purchase Department	230	230
Sales Department	485	485
Service Department	1332	1332
Grand Total	4697	4697

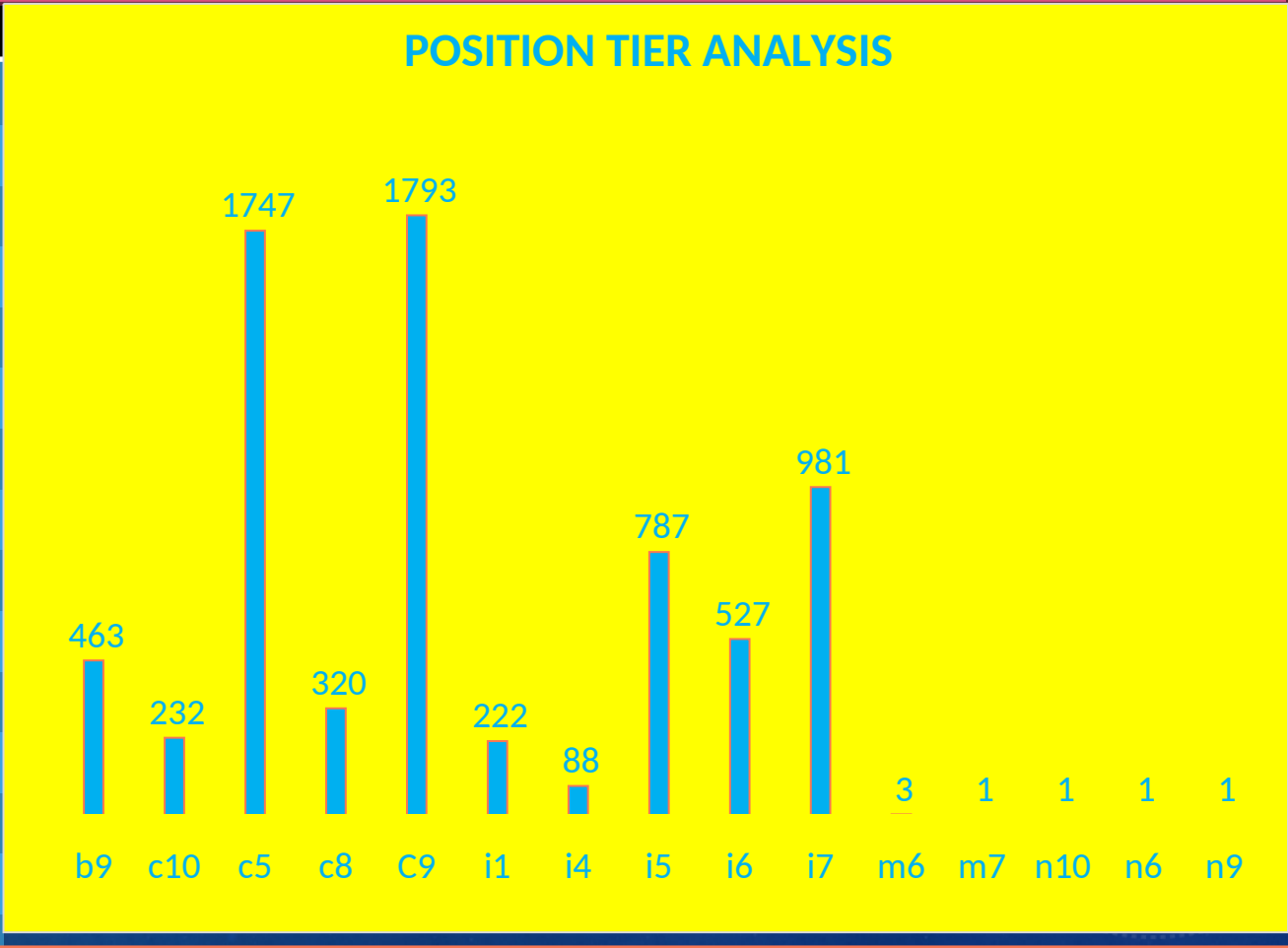


E. Position Tier Analysis:

Different positions within a company often have different tiers or levels.

Your Task: Use a chart or graph to represent the different position tiers within the company. This will help you understand the distribution of positions across different tiers.

Post Name	Count of Post Name
b9	463
c10	232
c5	1747
c8	320
C9	1793
i1	222
i4	88
i5	787
i6	527
i7	981
m6	3
m7	1
n10	1
n6	1
n9	1



RESULT:

Significant insights into the hiring process analytics and by analyzing hiring patterns, salary distribution, and departmental composition help optimize recruitment strategies and drive organizational growth

IMDB MOVIE ANALYSIS

PROJECT 5

The project was related to IMDB Movies.

The objective of this project is to conduct a comprehensive data analysis of the IMDB Movie datasets. The analysis mainly focuses on several key aspects such as Movie genres, Duration, Language, Directors and Budget and its impact on IMDB scores and financial success of the movies.

APPROACH:

DATA CLEANING:

DATA CLEANING STEPS

1 FULL DATASET INTO TABLE

2 REMOVING ALL BLANK ROWS

3 SEPARATED GENRES COLUMN

4 REMOVING DUPLICATES AND MAKE UNIQUE GENRE ITEMS

5 REPLACING NON PRINTABLE CHARACTER IN MOVIE TITLE COLUMN

6 REMOVING NON-NECESSARY COLUMNS FOR PROJECT

A. Movie Genre Analysis: Analyse the distribution of movie genres and their impact on the IMDB score.																							
Task: Determine the most common genres of movies in the dataset. Then, for each genre, calculate descriptive statistics (mean, median, mode, range, variance, standard deviation) of the IMDB scores.																							
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	
	Action	Adventure	Drama	Animation	Comedy	Mystery	Crime	Biography	Fantasy	Documentary	Sci-Fi	Horror	Romance	Family	Western	Musical	Thriller	History	Music	War	Sport	Film-Noir	
MEAN	6.3	6.4	6.8	6.7	6.2	6.5	6.5	7.2	6.3	7	6.3	5.9	6.4	6.2	6.8	6.6	6.4	7.2	6.5	7.1	6.6	7.7	
MODE	6.6	6.6	6.7	6.7	6.7	6.6	6.6	7	6.7	6.6	6.7	5.9	6.5	5.4	6	7.1	6.5	7.7	6.2	7.1	7.2		
MEDIAN	6.3	6.6	6.9	6.8	6.3	6.5	6.6	7.2	6.4	7.4	6.4	6	6.5	6.3	6.8	6.75	6.4	7.2	6.6	7.1	6.8	7.7	
MAX	9	8.9	9.3	8.6	8.8	8.6	9.3	8.9	8.9	8.5	8.8	8.6	8.5	8.6	8.9	8.5	9	8.9	8.5	8.6	8.3	7.7	
MIN	2.1	2.3	2.1	2.8	1.9	3.1	2.4	4.5	2.2	1.6	1.9	2.3	2.1	1.9	4.7	2.1	2.7	5.5	1.6	4.3	2	7.7	
STD DEV	1.04	1.11	0.9	0.99	1.04	1.02	0.98	0.69	1.13	1.38	1.16	1	0.95	1.16	0.93	1.1	0.97	0.67	1.18	0.8	1.04		
VAR	1.08	1.24	0.8	0.98	1.07	1.03	0.97	0.48	1.29	1.92	1.35	1	0.91	1.35	0.87	1.21	0.94	0.45	1.4	0.64	1.09		
E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z		
Action	Adventure	Drama	Animation	Comedy	Mystery	Crime	Biography	Fantasy	Documentary	Sci-Fi	Horror	Romance	Family	Western	Musical	Thriller	History	Music	War	Sport	Film-Noir		
7.9	7.9	6.7	7.8	7.8	7.5	7.2	5.5	7.9	7.8	7.9	7	6.2	7.8	6.5	7.8	6.8	6.7	7.8	6.1	6.3	7.7		
7.1	7.1	7.2	7.3	6.8	7.5	6.2	7.5	7.1	7.3	6.6	5.2	7.8	7.5	4.8	7.6	8.5	6.1	7.6	5.5	6.5			
6.8	6.8	7.7	6.3	7.3	6.5	9	6.8	7.8	7.1	7.5	7.2	7.2	6.6	8.1	5.9	5.9	5.5	5.9	7.7	6.1			
8.5	6.6	7.3	8.3	6.3	7.5	6.7	7	7.5	1.6	6.9	6.2	7.7	6.8	7.2	6.4	7	7.7	6.4	7.4	7.1			
6.6	6.2	6.8	7.2	8.3	7.6	6.1	7.8	7.3	5.9	6.1	5.8	7.3	6.1	5.4	7.1	7.8	7.4	5.4	6.6	6.8			
6.2	7.8	6.6	8.4	7.2	6.7	6.6	8.2	7.2	4.1	7.2	5.7	7.3	6.5	8.5	6.9	6.8	6.6	7.1	6	5.3			
7.5	7.5	9	6.8	6.2	6.1	6.6	7	6.6	8	8.1	6.6	6.6	7.3	6	7.5	7.2	7.1	6.9	7.1	6.8			
6.9	7.5	7.5	8.3	8.3	7.7	6.4	8	6.7	5.4	6.8	4.9	6.6	6.4	7	7	7.2	7.5	6.2	8				
6.1	6.9	8.3	6.5	6.5	8.1	7.3	8.2	6.8	7.5	8.2	5.7	7	6.3	5.9	4.4	8	7	5	7.2	7.3			
6.7	6.1	7.8	8.3	4.8	6.4	7.5	7.1	7.5	6.6	5.9	7	7.8	8.3	6.7	7.1	7.5	7	7	7.2	6.4			
7.3	6.7	6.1	6.4	6.9	7.4	4.8	8.4	7	2.7	7	4.9	6.1	7.2	6.7	5.8	6.2	7.6	4.4	6.9	6.3			
6.5	7.3	7.6	7.9	5.4	7.8	6.2	7	7.9	8.5	7.2	4.2	5.5	8.4	6.6	8.5	9	7.7	6.5	7.6	6.6			
7.2	6.5	6.3	7.8	8.3	6.6	6.4	7.9	6.1	7.6	6.8	5.1	6.6	6.8	7.9	6.7	5.2	6	7.1	7.7	4.5			
6.6	7.2	7.9	6.6	6.4	7.5	6.5	7.4	7.3	6.6	6	5.6	7.3	6.9	6.5	6.5	6.1	7.1	4.4	6	7.2			
8.1	6.6	8.6	8.2	7.9	7	5.8	7.2	6.5	8	5.7	6.2	6.4	8.3	6.6	5.9	5.8	6.7	7.3	7.5	3			
6.7	8.1	7.8	6.1	7.8	7	3.7	7.1	6.8	5.4	6.7	6.3	5.5	6.5	7.8	7.6	8.8	7.3	5.8	7.1	7.2			
6.8	6.7	5.6	8	6.1	5.8	6.5	6.6	7.3	7.2	6.8	5.2	5.5	7.5	4.7	5.2	6.7	7.6	8.5	6.7	7.1			
7	6.8	6.1	6.7	6.7	7.1	7	7	6.4	6.6	5.6	6.4	6.6	5.4	5.8	7	5.6	7.7	6.7	7.6	7.1			
6.7	7.5	5.5	7.6	8	7.7	7.8	6.6	6.7	5.1	6.6	5.9	5.3	8.3	6.8	6.6	6.7	7.2	6.5	7.2	3.8			
7.2	7	6.4	6.9	6.7	7.4	6.4	7.4	8.3	6.7	7	5.5	6.3	7.8	6.3	6.3	8.1	7.3	5.7	7.6	6.3			
8.2	6.7	7.2	5.1	5.9	6.1	6.7	6.7	8	7.7	8	5.9	6.5	7	7.4	7.4	6.7	5.6	4.5	6.6	5.9			
5.9	7.9	6.9	6.2	7.6	6.9	8.2	7.3	6.3	7.6	7.8	5.4	3.7	6.4	6.1	6.4	7.4	7.1	5.9	8.4	6.8			
7	6.1	5.8	7.3	6.9	5.7	3.3	7.9	6.6	7.1	7	6.8	7.1	6.5	5.2	6.4	5.8	7.6	7.6	7.1	7.4			
7.8	7.2	6.6	5.4	5.1	7.3	5.3	8	6.2	8	6.3	6.9	6.3	7.9	7.7	7.1	6.9	6.7	5.2	8.3	6.7			
7.3	8.2	6.4	6.7	6.2	6.3	7.5	6.3	7.2	8.1	7.5	5.7	4.4	7.8	6	7.6	6.4	8.4	7	7.1	6			
7.2	5.9	7.7	6.9	6.2	7.3	6.6	6.9	6.8	6.8	8.4	5.7	8.5	6.6	6.8	6.3	6.1	7.1	7.3	7.2	5.8			
6.8	7	6.1	6.9	7.3	6.6	5.5	7	6.9	7.8	5.8	6.7	7	8.2	6.4	6	6.6	7	6.7	6.1	5.8			
6	7.8	7.7	6.3	5.4	7.8	8.5	8	5.2	7.4	5.4	4.8	5.5	6.1	7.5	7.1	7.4	7.2	7.9	6.3	5.9			
5.7	7.3	7.4	7.2	6.7	7.6	7.8	7.4	5.4	7.7	7.3	6	6.9	6.4	7.5	7.4	5	7.4	5.9	7	7.3			
6.7	7.2	5.5	7.3	6.9	4.9	7.6	8.2	8.3	8.2	6.5	5.6	6	6.1	6	6.4	6.6	7.6	5.2	6.3	7.6			
6.8	6.5	8.1	7.9	6.9	7.4	6.4	6.7	7.8	7.7	7.9	6.4	3.3	6.3	7.5	7.5	5.8	6	6.6	5.4	4.7			
5.6	6.8	5.5	6.4	5.5	6.4	6.9	7.3	6.1	7.3	4.8	5.9	7.6	7.5	7.2	6.9	6	7.1	6.3	8.6	6.5			
6.6	7.3	5.8	5.9	6.1	6.9	7	7.6	7	7.6	6.9	6.4	4.3	7.6	7.8	7.2	6.4	6.1	7.4	7.4	6.5			
7.2	6	6.6	6.5	7.2	7.8	6.1	6.7	6.3	8.3	7.9	6.4	6.9	6.7	5.8	7	8.1	7.4	6.4	7.2	7.2			
7	5.7	6.4	7.2	6.1	6.8	6.7	6.4	6.4	8.1	6.1	6.8	7	8	6	6.2	8.1	6.6	6.4	6.6	6.3			

B. Movie Duration Analysis:

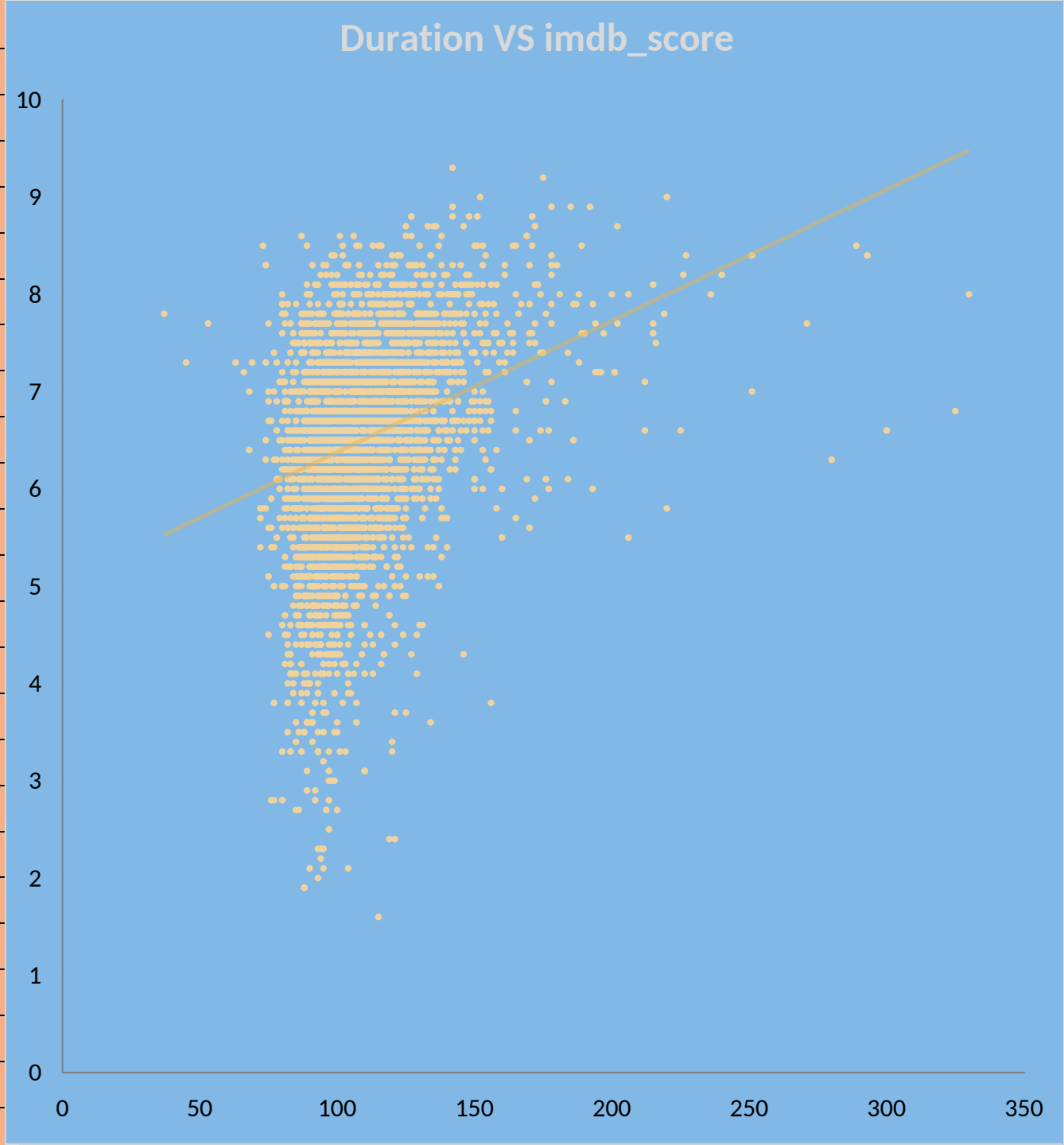
Analyse the distribution of movie durations and its impact on the IMDB score.

Task: :

Analyse the distribution of movie durations and identify the relationship between movie duration and IMDB score.

MOVIE DURATION	
MEAN	110.26
MEDIA	106.00
STD	22.65

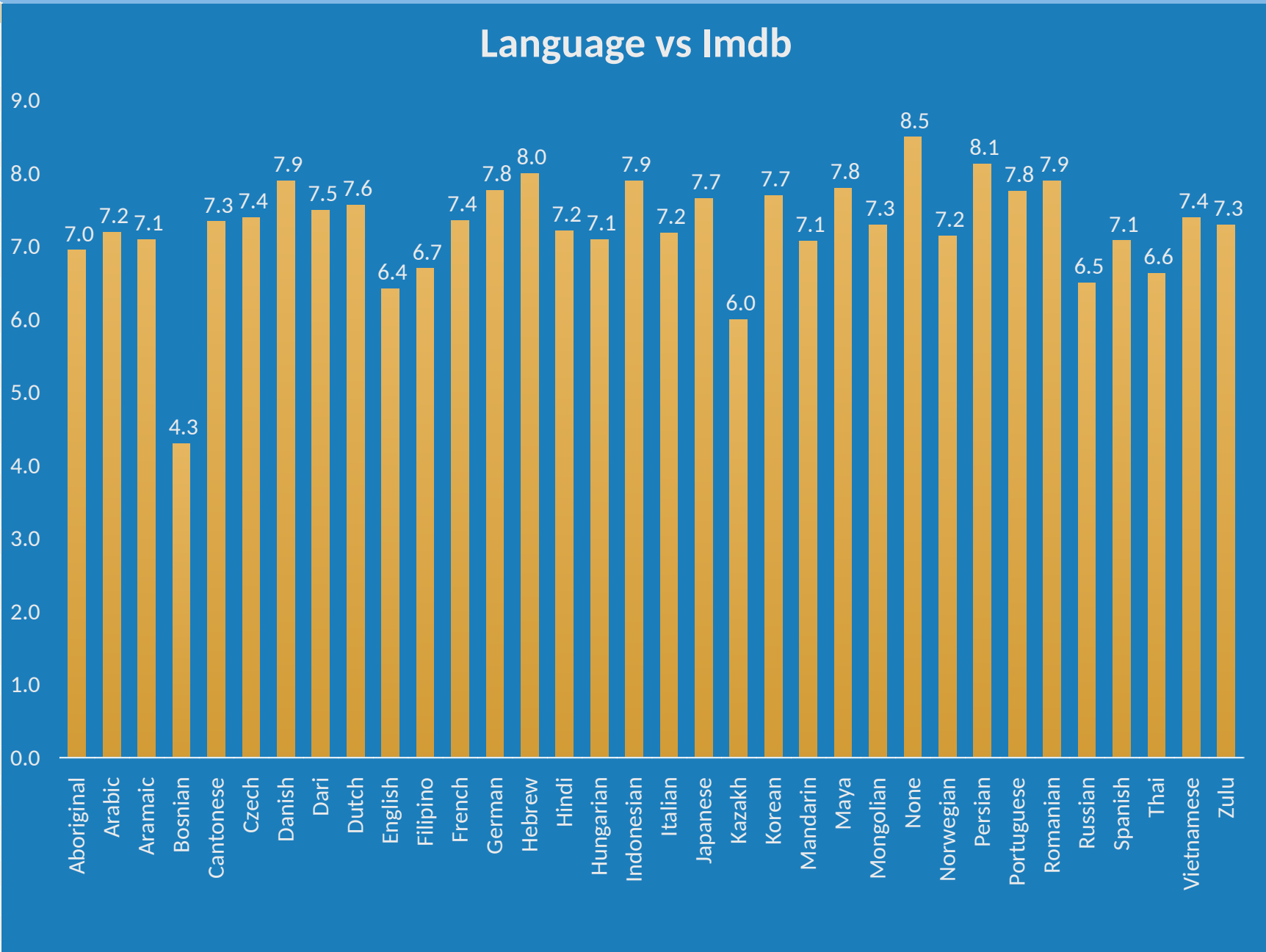
duration	imdb_score
178	7.9
169	7.1
148	6.8
164	8.5
132	6.6
156	6.2
100	7.8
141	7.5
153	7.5
183	6.9
169	6.1
106	6.7
151	7.3
150	6.5
143	7.2
150	6.6
173	8.1
136	6.7
106	6.8
164	7.5
153	7
156	6.7
186	7.9
113	6.1



C. Language Analysis: Situation: Examine the distribution of movies based on their language.

Task: Determine the most common languages used in movies and analyse their impact on the IMDB score using descriptive statistics.

Language	Average of imdb_score
Aboriginal	7.0
Arabic	7.2
Aramaic	7.1
Bosnian	4.3
Cantonese	7.3
Czech	7.4
Danish	7.9
Dari	7.5
Dutch	7.6
English	6.4
Filipino	6.7
French	7.4
German	7.8
Hebrew	8.0
Hindi	7.2
Hungarian	7.1
Indonesian	7.9
Italian	7.2
Japanese	7.7
Kazakh	6.0
Korean	7.7
Mandarin	7.1
Maya	7.8
Mongolian	7.3
None	8.5
Norwegian	7.2
Persian	8.1
Portuguese	7.8
Romanian	7.9
Russian	6.5
Spanish	7.1
Thai	6.6
Vietnamese	7.4



Top 5 most common languages

Language	No of Movies
English	3598
French	34
Spanish	23
Mandarin	15
Japanese	10

D. Director Analysis: Influence of directors on movie ratings.

Task: Identify the top directors based on their average IMDB score and analyse their contribution to the success of movies using percentile calculations.

director_name	Average of imdb_score	percentaile
Akira Kurosawa	8.7	99.40%
Tony Kaye	8.6	99.20%
Charles Chaplin	8.6	99.20%
Ron Fricke	8.5	98.70%
Majid Majidi	8.5	98.70%
Damien Chazelle	8.5	98.70%
Alfred Hitchcock	8.5	98.70%
Sergio Leone	8.4	98.70%
Christopher Nolan	8.4	98.70%
Richard Marquand	8.4	98.30%

TOP 10 DIRECTOR VS imdb_score

8.7



8.6



8.6



8.5



8.5



8.5



8.5



8.4



8.4



8.4



Akira
Kurosawa

Tony Kaye

Charles
Chaplin

Ron Fricke

Majid Majidi

Damien
Chazelle

Alfred
Hitchcock

Sergio Leone

Christopher
Nolan

Richard
Marquand

E. Budget Analysis: Explore the relationship between movie budgets and their financial success.

gross	PROFIT MARGIN	CORRELATION	MAX	MIN	E) BUDGET ANALYSIS							
760505847	523505847	0.10	523505847	-12213298588								
309404152	9404152											
200074175	-44925825				B			C			D	
448130642	198130642				budget			gross			PROFIT MARGIN	
73058679	-190641321				237000000			760505847			523505847	
336530303	78530303				220000000			623279547			403279547	
200807262	-59192738				200000000			658672302			458672302	
458991599	208991599				150000000			652177271			502177271	
301956980	51956980				185000000			533316061			348316061	
330249062	80249062				115000000			474544677			359544677	
200069408	-8930592				78000000			407999255			329999255	
168368427	-31631573				45000000			422783777			377783777	
423032628	198032628				220000000			623279547			403279547	
89289910	-125710090				11000000			460935665			449935665	
291021565	66021565				10500000			434949459			424449459	
141614023	-83385977											
623279547	403279547											
241063875	-8936125											
179020854	-45979146											
255108370	5108370											
262030663	32030663											
105219735	-94780265											
258355354	33355354											
70083519	-109916481											

BANK LOAN CASE-STUDY

PROJECT 6

PROJECT DESCRIPTION:

The case study aims to identify a pattern if a client has difficulty paying their installment. This project also provides an example of how EDA (EXPLORATORY DATA ANALYSIS) might be used in the real-world corporate setting. Some learning about risk analytics in banking. The goal is to detect what are the trends that show whether a client has trouble paying their installments. The process starts from data cleaning to the basic method of data analysis such as univariate, segment univariate and bivariate analysis and correlation between variables

APPROACH:

To perform the project a systematic approach was followed. A dataset was downloaded. Microsoft Excel 2021 was selected as the primary tool for data analysis such as cleaning, missing data and other EDAs as needed due to its versatility and robust capabilities in handling tabular data. Specific techniques such as pivot tables, charts, and formulas were employed to analyze the dataset and extract meaningful insights.

TECH-STACK USED:

Software: Microsoft Excel 2021

Purpose: Excel was chosen for its extensive data analysis functions, including pivot tables, charts, and statistical functions and visualization of the hiring data

A. IDENTIFY MISSING DATA AND DEAL WITH IT APPROPRIATELY:
As a data analyst, you come across missing data in the loan application dataset. It is essential to handle missing data effectively to ensure the accuracy of the analysis.

Task:
Identify the missing data in the dataset and decide on an appropriate method to deal with it using Excel built-in functions and features.

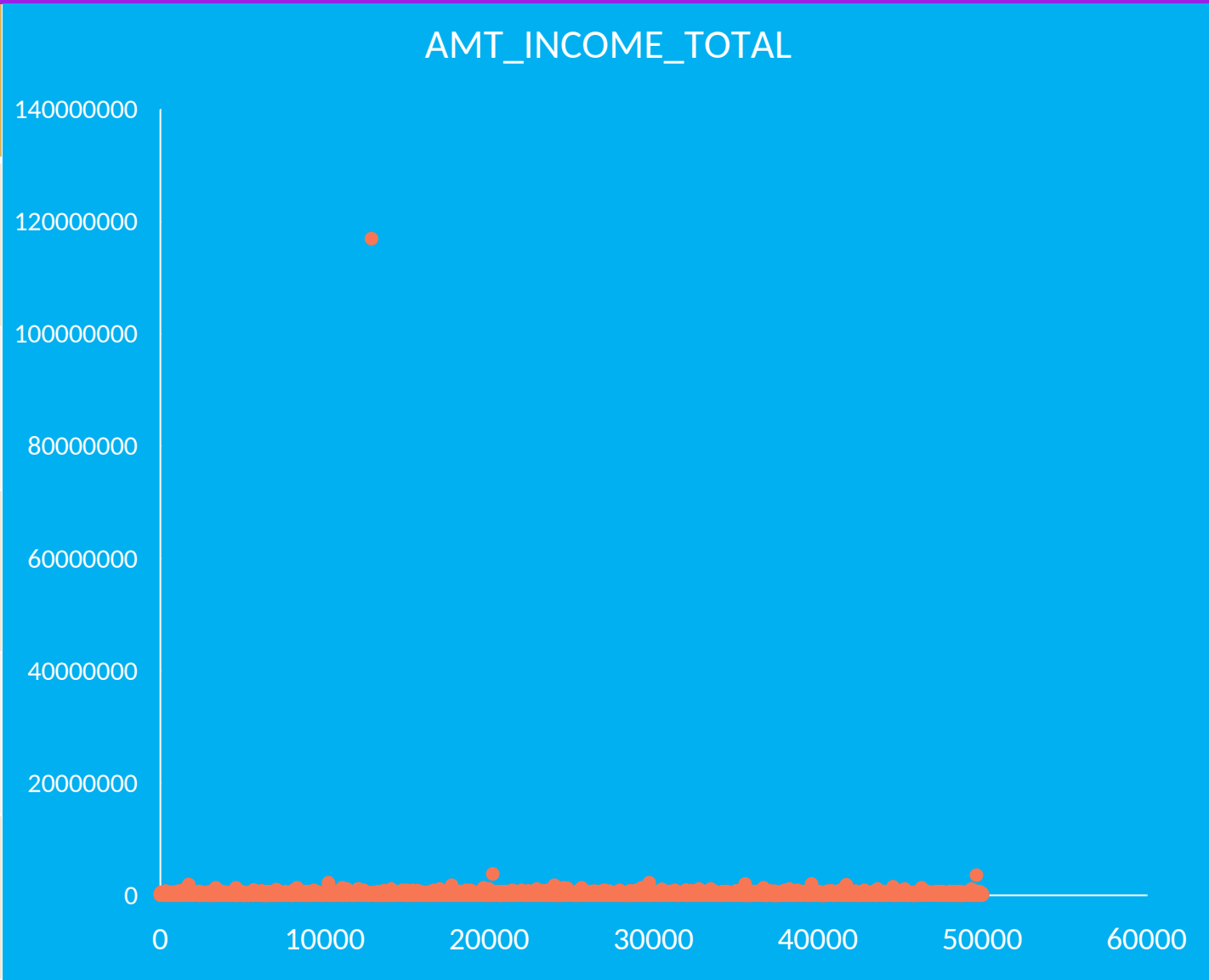
PROPORTION MISSING VALUE		MODIFIED METHOD	
< 50%		FOR NUMERICAL VALUES AVERAGE AND MEDIAN WILL BE PREFERRED FOR CATEGORICAL VALUES MODE WILL BE PREFERRED	
>50		REMOVE THE WHOLE COLUMN COMPLETELY	
SL.NO	REPLACING MISSING COLUMNS	FORMULA(AVERAGE)	MEDIAN
1	AVERAGE AMT_ANNUITY	27107.38	24939
2	AVERAGE AMT_GOODS_PRICE	539060.04	450000
3	NAME_TYPE_SUITE	Unaccompanied	MODE
4	OCCUPATION_TYPE	Laborers	MODE
5	AVERAGE CNT_FAM_MEMBERS	2.16	2
6	OBS_30_CNT_SOCIAL_CIRCLE	1.42	0
7	DEF_30_CNT_SOCIAL_CIRCLE	0.14	0
8	OBS_60_CNT_SOCIAL_CIRCLE	1.40	0
9	DEF_60_CNT_SOCIAL_CIRCLE	0.10	0
10	DAYS_LAST_PHONE_CHANGE	-964	-755
11	AMT_REQ_CREDIT_BUREAU_HOUR	0	0
12	AMT_REQ_CREDIT_BUREAU_DAY	0	0
13	AMT_REQ_CREDIT_BUREAU_WEEK	0	0
14	AMT_REQ_CREDIT_BUREAU_MON	0	0
15	AMT_REQ_CREDIT_BUREAU_QRT	0	0
16	AMT_REQ_CREDIT_BUREAU_YEAR	2	2

B. Identify Outliers in the Dataset:

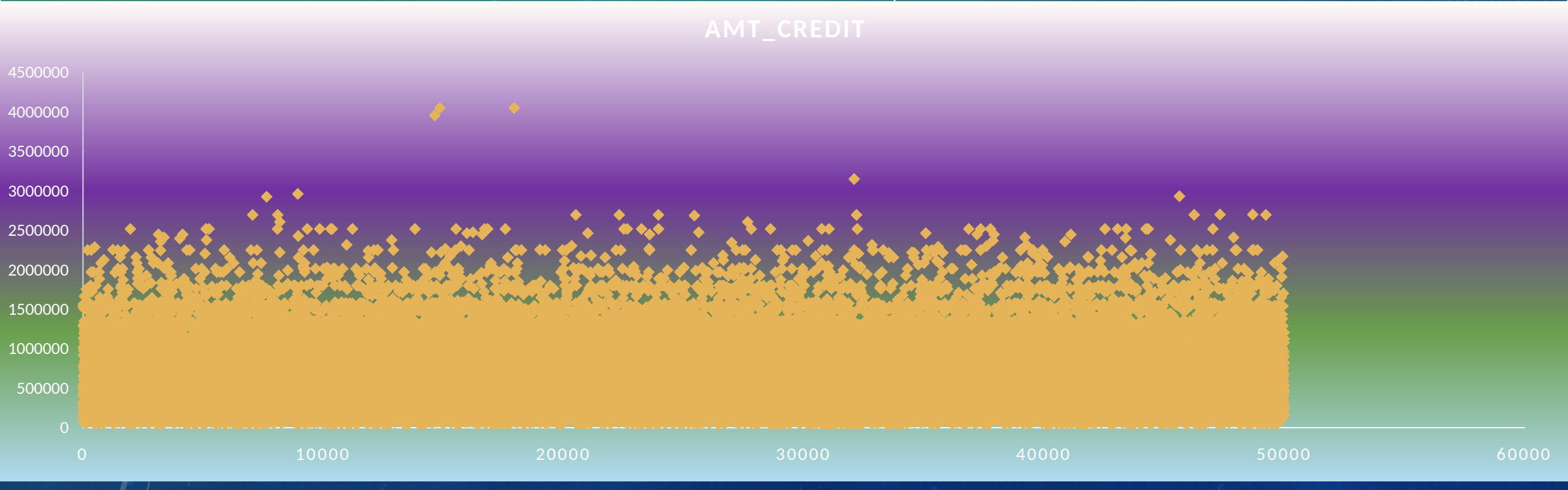
Outliers can significantly impact the analysis and distort the results. You need to identify outliers in the loan application dataset

Task: Detect and identify outliers in the dataset using Excel statistical functions and features, focusing on numerical variables.

Q1	112500
Q3	202500
IQR(Q3-Q1)	90000
MIN	-22500
MAX	225000
MEDIAN	145800



Q1	270000
Q3	808650
IQR(Q3-Q1)	538650
MIN	-537975
MAX	1616625
MEDIAN	514777.5

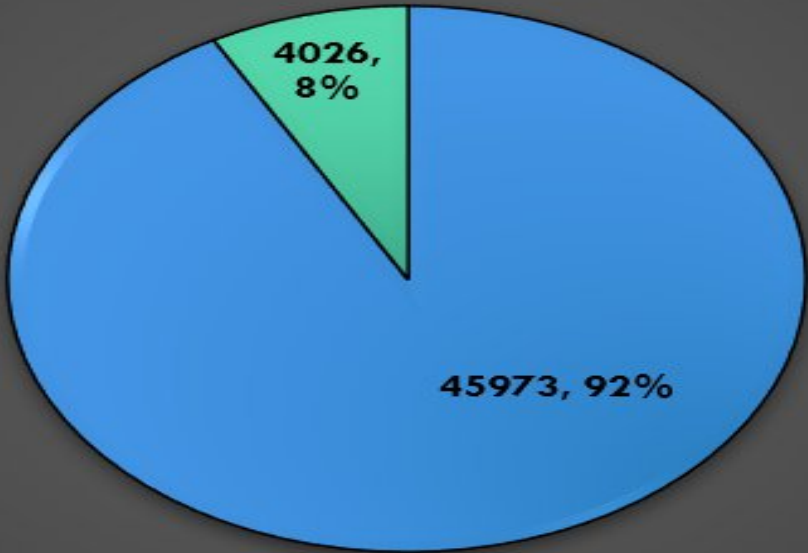


C. Analyse Data Imbalance: Data imbalance can affect the accuracy of the analysis, especially for binary classification problems. Understanding the data distribution is crucial for building reliable models.

Task: Determine if there is data imbalance in the loan application dataset and calculate the ratio of data imbalance using Excel functions.

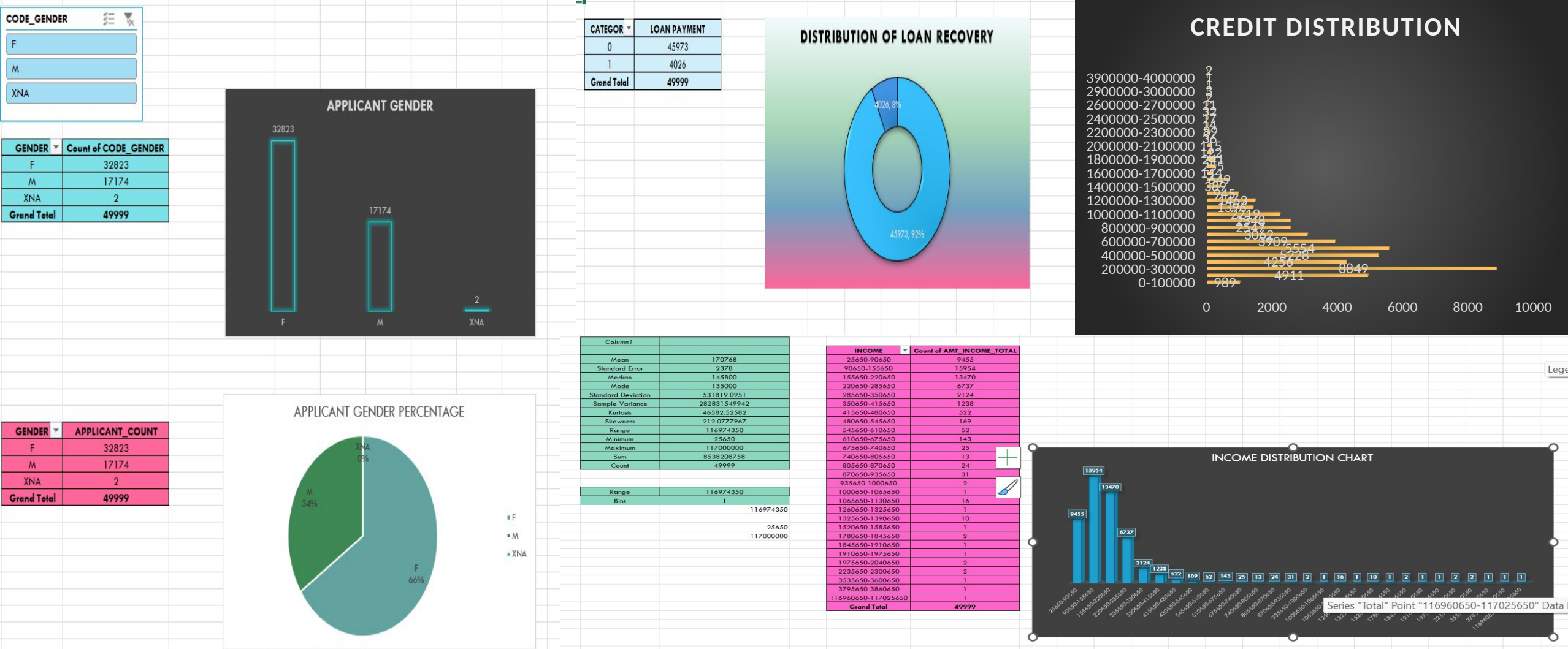
TARGET	Count of TARGET	TARGET		
0	45973	0	NO PAYMENT DIFFICULTY	45973
1	4026	1	DEFAULT CHANCE	4026
Grand Total	49999		RATIO OF DATA IMBALANCE	11.42

DATA IMBALANCE IN TARGET



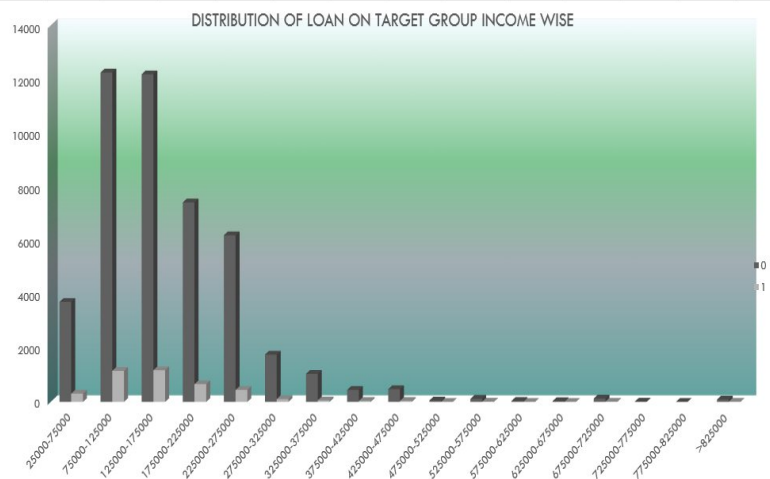
D. Perform Univariate, Segmented Univariate, and Bivariate Analysis: To gain insights into the driving factors of loan default, it is important to conduct various analyses on consumer and loan attributes.

Task: Perform univariate analysis to understand the distribution of individual variables, segmented univariate analysis to compare variable distributions for different scenarios, and bivariate analysis to explore relationships between variables and the target variable using Excel functions and features.



SEGMENT UNIVARIATE ANALYSIS:

Count of TARGET	TARGET GROUP ▾		
INCOME AMOUNT ▾	0	1	Grand Total
5000-75000	3721	309	4030
5000-125000	12254	1156	13410
25000-175000	12186	1179	13365
75000-225000	7421	661	8082
25000-275000	6198	447	6645
75000-325000	1757	104	1861
25000-375000	1045	58	1103
75000-425000	449	40	489
25000-475000	475	38	513
75000-525000	53	3	56
25000-575000	116	10	126
75000-625000	39	4	43
25000-675000	29	1	30
75000-725000	126	9	135
25000-775000	12		12
75000-825000	2		2
825000	90	7	97
Grand Total	45973	4026	49999
</			



Count of TARGET	LOAN DEFAULTERS	
CONTRACT	1	Grand Total
Cash loans	3792	3792
Revolving loans	234	234
Grand Total	4026	4026

Count of TARGET	LOAN REGULAR PAYERS	
CONTRACT TYPE	0	Grand Total
Cash loans	41484	41484
Revolving loans	4489	4489
Grand Total	45973	45973

Count of TARGET	LOAN REGULAR PAYERS	
CONTRACT TYPE	0	
Cash loans	41484	
Revolving loans	4489	
Grand Total	45973	

0 LOAN REGU
1 DEFAULTERS

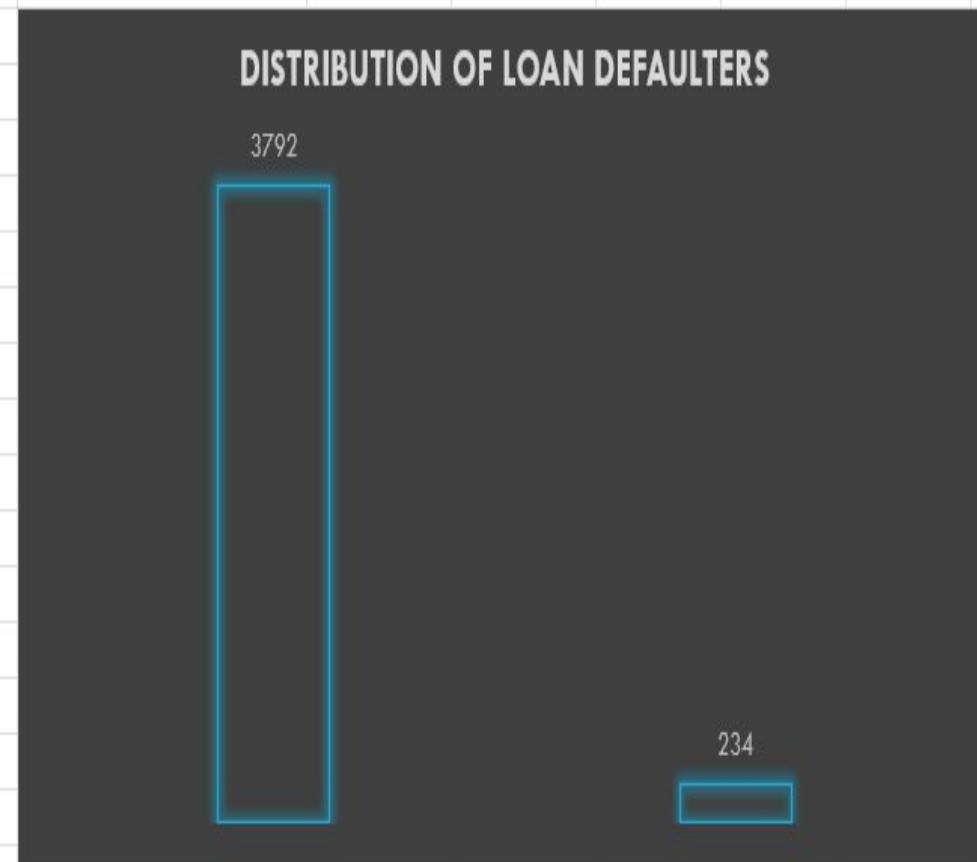
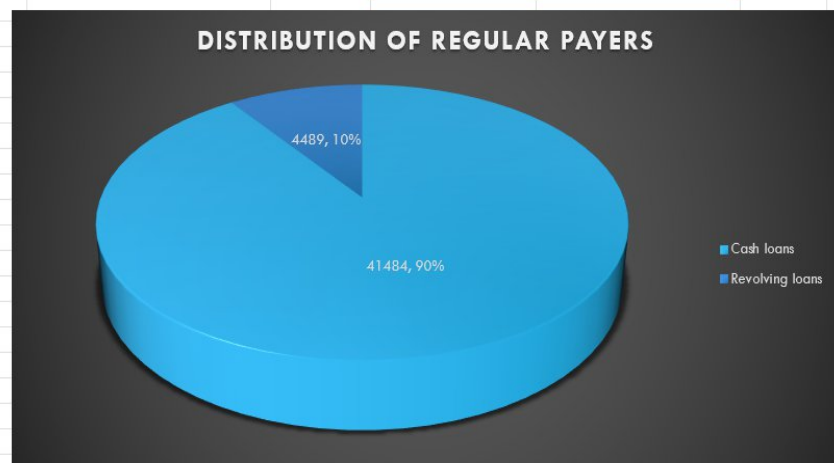
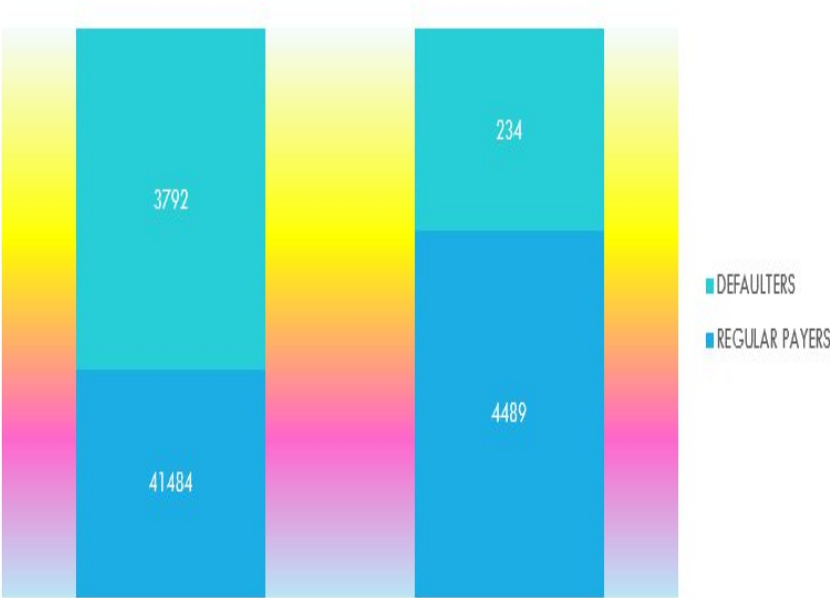


Chart Area

BIVARIATE ANALYSIS:

A	B	C	D	E	F	G	H	I
Count of TARGET	TARGET GROUP							
CONTRACT TYPE	REGULAR PAYERS	DEFAULTERS	Grand Total		CONTRACT TYPE	REGULAR PAYERS	DEFAULTERS	
Cash loans	41484	3792	45276		Cash loans	41484	3792	
Revolving loans	4489	234	4723		Revolving loans	4489	234	
Grand Total	45973	4026	49999		Grand Total	45973	4026	

LOAN CATEGORY AGAINST TARGET



0 REGULAR PAYERS

1 EFAULTERS

INCOME AMOUNT	Average of AMT_INCOME_TOTAL
25000-75000	60893.22
75000-125000	101124.41
125000-175000	145135.07
175000-225000	190600.10
225000-275000	241976.08
275000-325000	305219.24
325000-375000	352801.16
375000-425000	396784.97
425000-475000	447067.15
475000-525000	496896.43
525000-575000	541946.43
575000-625000	590860.47
625000-675000	630750.00
675000-725000	681650.00
725000-775000	758473.13
775000-825000	785250.00
>825000	2319355.67
Grand Total	17076759

AVERAGE INCOME AMOUNT

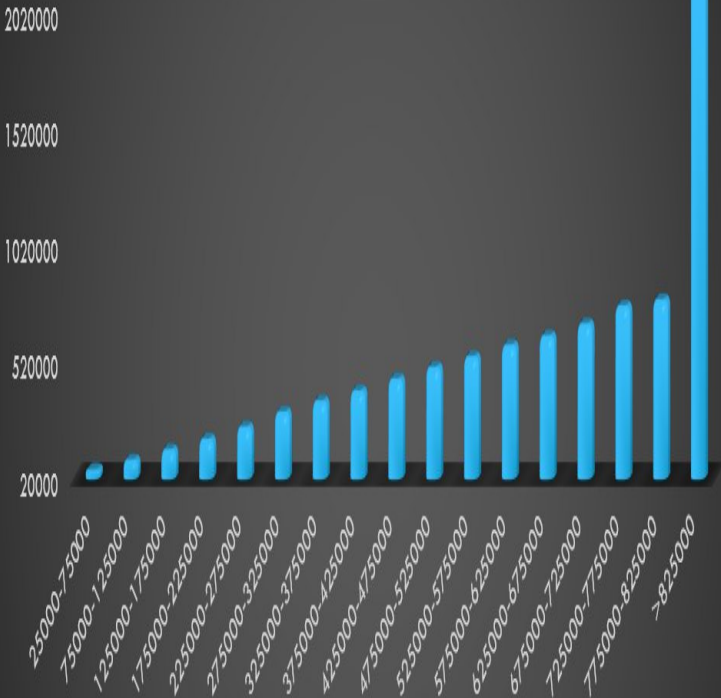
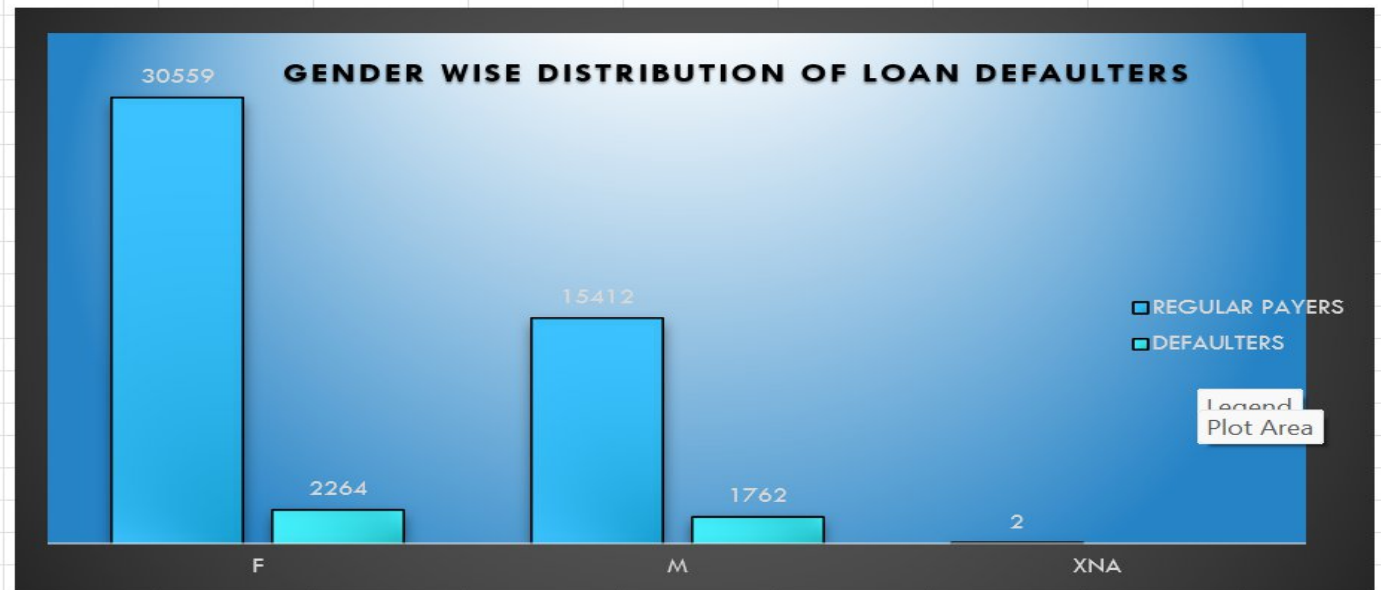
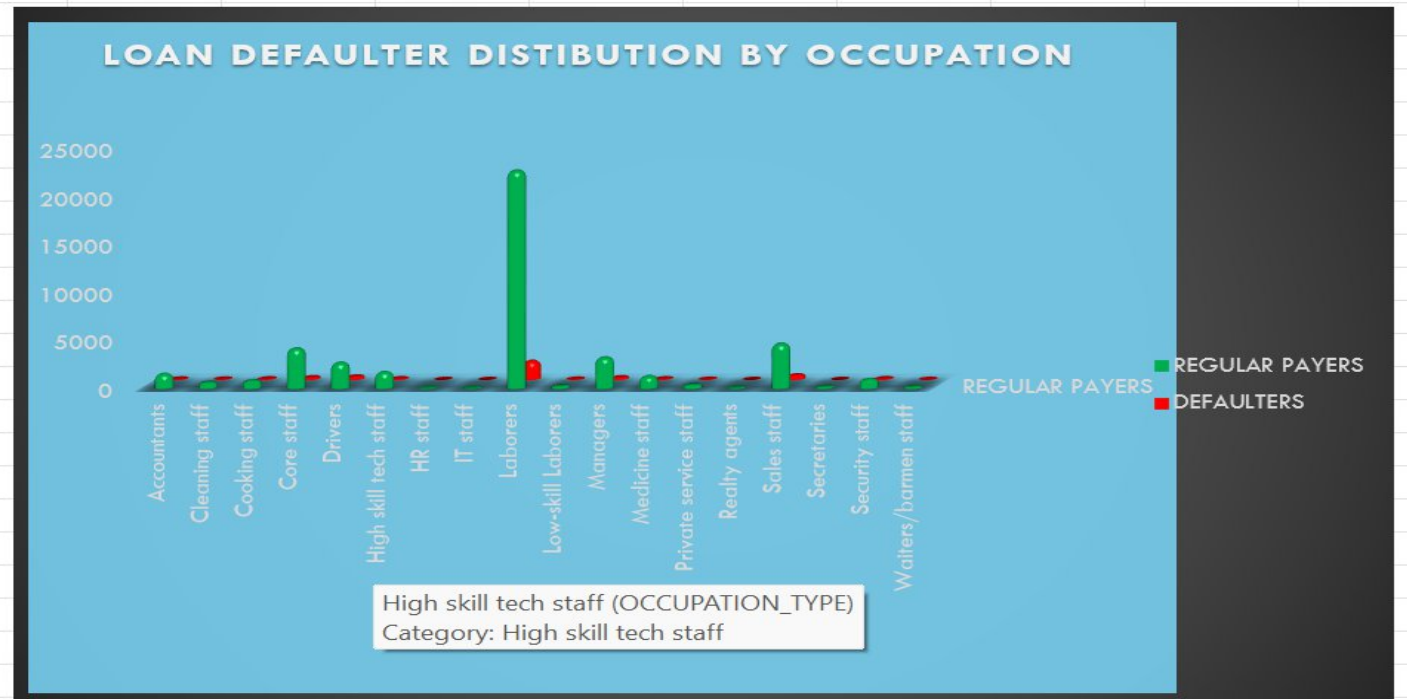


Chart Area

Count of TARGET	TARGET GROUP 		
GENDER 	REGULAR PAYERS	DEFAULTERS	Grand Total
F	30559	2264	32823
M	15412	1762	17174
XNA	2		2
Grand Total	45973	4026	49999



Count of TARGET	TARGET GROUP 		
OCCUPATION 	REGULAR PAYERS	DEFAULTERS	Grand Total
Accountants	1540	81	1621
Cleaning staff	671	68	739
Cooking staff	862	101	963
Core staff	4184	250	4434
Drivers	2706	338	3044
High skill tech staff	1734	118	1852
HR staff	92	9	101
IT staff	76	4	80
Laborers	22660	1946	24606
Low-skill Laborers	296	61	357
Managers	3246	243	3489
Medicine staff	1297	106	1403
Private service staff	410	37	447
Realty agents	110	13	123
Sales staff	4668	492	5160
Secretaries	203	9	212
Security staff	1015	125	1140
Waiters/barmen staff	203	25	228
Grand Total	45973	4026	49999



E. Identify Top Correlations for Different Scenarios: Understanding the correlation between variables and the target variable can provide insights into strong indicators of loan default.

Task: Segment the dataset based on different scenarios (e.g., clients with payment difficulties and all other cases) and identify the top correlations for each segmented data using Excel functions.

TARGET (0):

CNT_CHILDREN	AMT_INCOME_TOTAL	AMT_CREDIT	AMT_GOODS_PRICE	REGION_POPULATION_RELATIVE	DAYS_BIRTH	DAYS_EMPLOYED	DAYS_ID_PUBLISH	REGION_RATING_CLIENT	AMT_REQ_CREDIT_BUREAU_YEAR
1.00	0.04	0.01	0.00	-0.02	-0.34	-0.25	0.03	0.02	-0.04
0.04	1.00	0.38	0.38	0.18	-0.07	-0.16	-0.03	-0.21	0.02
0.01	0.38	1.00	0.99	0.10	0.05	-0.07	0.01	-0.10	-0.04
0.00	0.38	0.99	1.00	0.10	0.05	-0.07	0.01	-0.10	-0.05
-0.02	0.18	0.10	0.10	1.00	0.03	-0.01	0.00	-0.54	0.01
-0.34	-0.07	0.05	0.05	0.03	1.00	0.62	0.27	-0.01	0.06
-0.25	-0.16	-0.07	-0.07	-0.01	0.62	1.00	0.27	0.04	0.05
0.03	-0.03	0.01	0.01	0.00	0.27	0.27	1.00	0.01	0.03
0.02	-0.21	-0.10	-0.10	-0.54	-0.01	0.04	0.01	1.00	0.01
-0.04	0.02	-0.04	-0.05	0.01	0.06	0.05	0.03	0.01	1.00

TARGET (0):

AMT_INCOME_TOTAL	AMT_CREDIT	AMT_GOODS_PRICE	REGION_POPULATION_RELATIVE	DAYS_BIRTH	DAYS_EMPLOYED	DAYS_ID_PUBLISH	REGION_RATING_CLIENT	AMT_REQ_CREDIT_BUREAU_YEAR
0.01	0.01	0.00	-0.02	-0.25	-0.19	0.04	0.06	-0.04
1.00	0.02	0.01	-0.01	-0.01	-0.01	0.01	-0.01	-0.01
0.02	1.00	0.98	0.07	0.14	0.02	0.04	-0.05	-0.03
0.01	0.98	1.00	0.08	-0.14	0.14	0.02	0.02	-0.05
-0.01	0.07	0.08	1.00	0.02	0.01	0.01	-0.43	0.02
-0.01	0.14	0.14	0.02	1.00	0.59	0.25	-0.05	0.09
-0.01	0.02	0.02	0.01	0.59	1.00	0.23	-0.01	0.02
0.01	0.04	0.05	0.01	0.25	0.23	1.00	-0.03	0.06
-0.01	-0.05	-0.05	-0.43	-0.05	-0.01	-0.03	1.00	0.02
-0.01	-0.03	-0.03	0.02	0.09	0.02	0.06	0.02	1.00

TOP 5 CORRELATION (TAGET 0)- LOAN PAYERS

VARIABLE 1	VARIABLE 2	CORRELATION
DAYS_EMPLOYED	DAYS_BIRTH	0.62
AMT_CREDIT	AMT_INCOME_TOTAL	0.38
AMT_GOODS_PRICE	AMT_INCOME_TOTAL	0.27
DAYS_ID_PUBLISH	DAYS_BIRTH	0.27
DAYS_ID_PUBLISH	DAYS_EMPLOYED	0.27

TOP 5 CORRELATION (TARGET 1)- DEFAULTERS

VARIABLE 1	VARIABLE 2	CORRELATION
AMT_GOODS_PRICE	AMT_CREDIT	0.98
DAYS_BIRTH	DAYS_EMPLOYED	0.59
DAYS_ID_PUBLISH	DAYS_BIRTH	0.25
DAYS_ID_PUBLISH	DAYS_EMPLOYED	0.23
DAYS_BIRTH	AMT_GOODS_PRICE	0.14

RESULTS: The Bank Lona case study was executed starting with data cleaning process as missing data, identifying outliers and other exploratory data analysis then followed by different analysis and then identify the correlation between different variables

IMPACT OF CAR FEATURES

PROJECT 7

PROJECT DESCRIPTION:

- For the given dataset, as a Data Analyst, the client has asked How can a car manufacturer optimize pricing and product development decisions to maximize profitability while meeting consumer demand?
- This problem could be approached by analysing the relationship between a car's features, market category, and pricing, and identifying which features and categories are most popular among consumers and most profitable for the manufacturer.
- By using data analysis techniques such as regression analysis and market segmentation, the manufacturer could develop a pricing strategy that balances consumer demand with profitability, and identify which product features to focus on in future product development efforts

APPROACH:

- To perform the project a systematic approach was followed. A dataset was downloaded. Microsoft Excel 2021 was selected as the primary tool for data analysis such as cleaning, deleting blank rows and due to its versatility and robust capabilities in handling tabular data. Specific techniques such as pivot tables, charts, visualizations and were employed to analyze the dataset and extract meaningful insights.

TECH-STACK USED:

- Software: Microsoft Excel 2021
- Purpose: Excel was chosen for its extensive data analysis functions, including pivot tables, charts, and statistical functions and visualization of the Car data

TASK ANALYSIS:

Insight Required: How does the popularity of a car model vary across different market categories?

- Task 1.A: Create a pivot table that shows the number of car models in each market category and their corresponding popularity scores.
- Task 1.B: Create a combo chart that visualizes the relationship between market category and popularity.

MARKET CATEGORY	Average of Popularity	Count of Model	Exotic,Luxury,High-Performance	467.08	79
Crossover	1529.03	1103	Exotic,Luxury,High-Performance,Hybrid	204.00	1
Crossover,Diesel	873.00	7	Exotic,Luxury,Performance	217.03	36
Crossover,Exotic,Luxury,High-Performance	238.00	1	Factory Tuner,High-Performance	1941.42	106
Crossover,Exotic,Luxury,Performance	238.00	1	Factory Tuner,Luxury	617.00	2
Crossover,Factory Tuner,Luxury,High-Performance	1823.46	26	Factory Tuner,Luxury,High-Performance	2133.37	215
Crossover,Factory Tuner,Luxury,Performance	2607.40	5	Factory Tuner,Luxury,Performance	1413.42	31
Crossover,Factory Tuner,Performance	210.00	4	Factory Tuner,Performance	1733.10	89
Crossover,Flex Fuel	2073.75	64	Flex Fuel	2217.30	872
Crossover,Flex Fuel,Luxury	1173.20	10	Flex Fuel,Diesel	5657.00	16
Crossover,Flex Fuel,Luxury,Performance	1624.00	6	Flex Fuel,Factory Tuner,Luxury,High-Performance	258.00	1
Crossover,Flex Fuel,Performance	5657.00	6	Flex Fuel,Hybrid	155.00	2
Crossover,Hatchback	1675.69	72	Flex Fuel,Luxury	746.54	39
Crossover,Hatchback,Factory Tuner,Performance	2009.00	6	Flex Fuel,Luxury,High-Performance	878.91	33
Crossover,Hatchback,Luxury	204.00	7	Flex Fuel,Luxury,Performance	1380.07	28
Crossover,Hatchback,Performance	2009.00	6	Flex Fuel,Performance	1702.36	81
Crossover,Hybrid	2563.38	42	Flex Fuel,Performance,Hybrid	155.00	2
Crossover,Luxury	884.55	410	Hatchback	1293.00	614
Crossover,Luxury,Diesel	2195.85	33	Hatchback,Diesel	873.00	14
Crossover,Luxury,High-Performance	1037.22	9	Hatchback,Factory Tuner,High-Performance	1205.15	13
Crossover,Luxury,Hybrid	630.92	24	Hatchback,Factory Tuner,Luxury,Performance	886.89	9
Crossover,Luxury,Performance	1344.85	113	Hatchback,Factory Tuner,Performance	2159.05	22
Crossover,Luxury,Performance,Hybrid	3916.00	2	Hatchback,Flex Fuel	5657.00	7
Crossover,Performance	2585.96	69	Hatchback,Hybrid	2121.25	72
Diesel	1730.90	84	Hatchback,Luxury	1379.50	46
Diesel,Luxury	2275.00	51	Hatchback,Luxury,Hybrid	454.00	3
Exotic,Factory Tuner,High-Performance	1046.38	21	Hatchback,Luxury,Performance	1566.13	38
Exotic,Factory Tuner,Luxury,High-Performance	517.54	52	Hatchback,Performance	1039.65	252
Exotic,Factory Tuner,Luxury,Performance	520.00	3	High-Performance	1821.45	199
Exotic,Flex Fuel,Factory Tuner,Luxury,High-Performance	520.00	13	Hybrid	2105.57	123
Exotic,Flex Fuel,Luxury,High-Performance	520.00	11	Luxury	1107.55	851
			Luxury,High-Performance	1668.02	334
			Luxury,High-Performance,Hybrid	568.83	12
			Luxury,Hybrid	724.69	48
			Luxury,Performance	1292.62	673
			Luxury,Performance,Hybrid	2333.18	11
			Performance	1371.08	584
			Performance,Hybrid	155.00	1
			Grand Total	1499.397823	8084

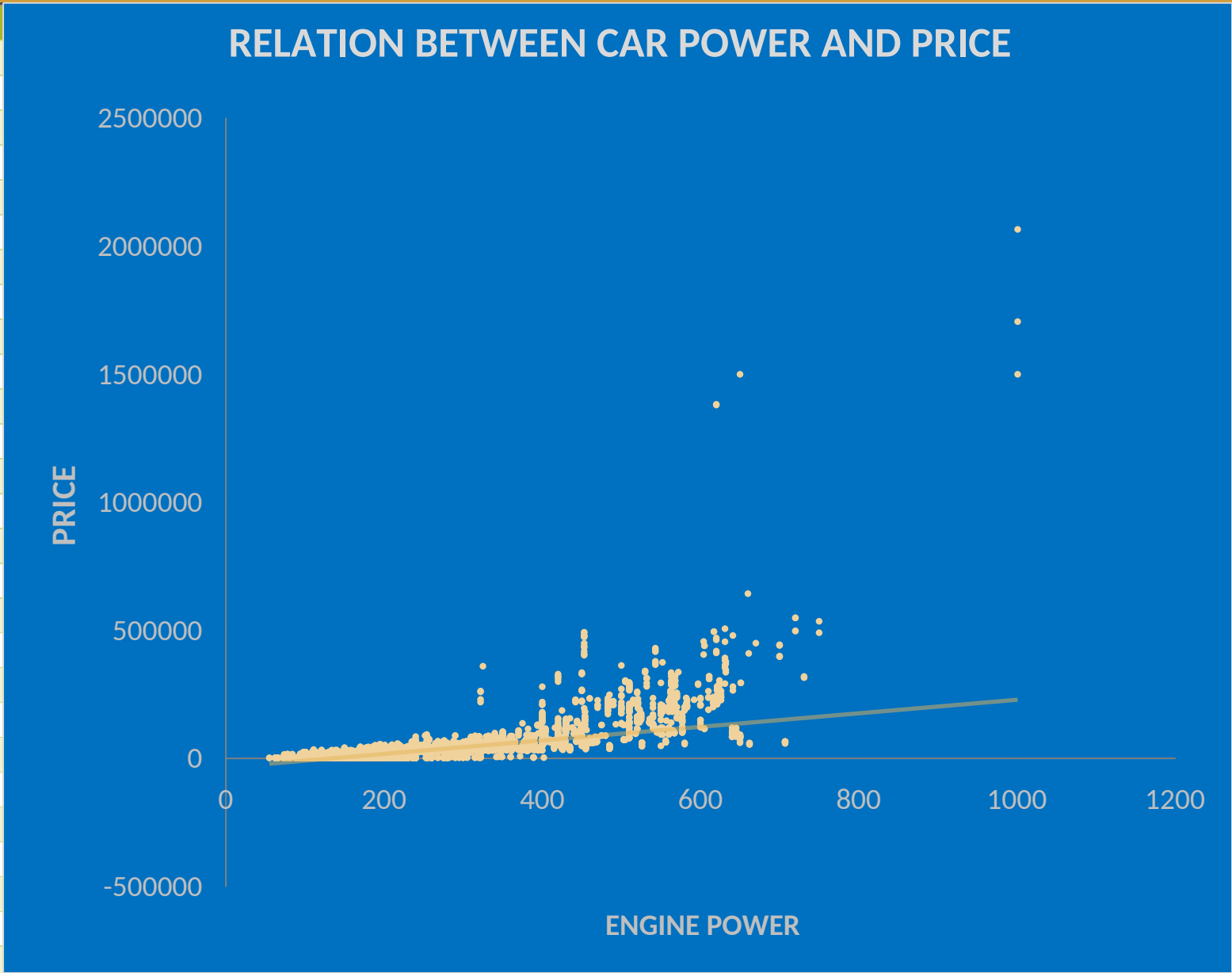
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Insight Required: What is the relationship between a car's engine power and its price?

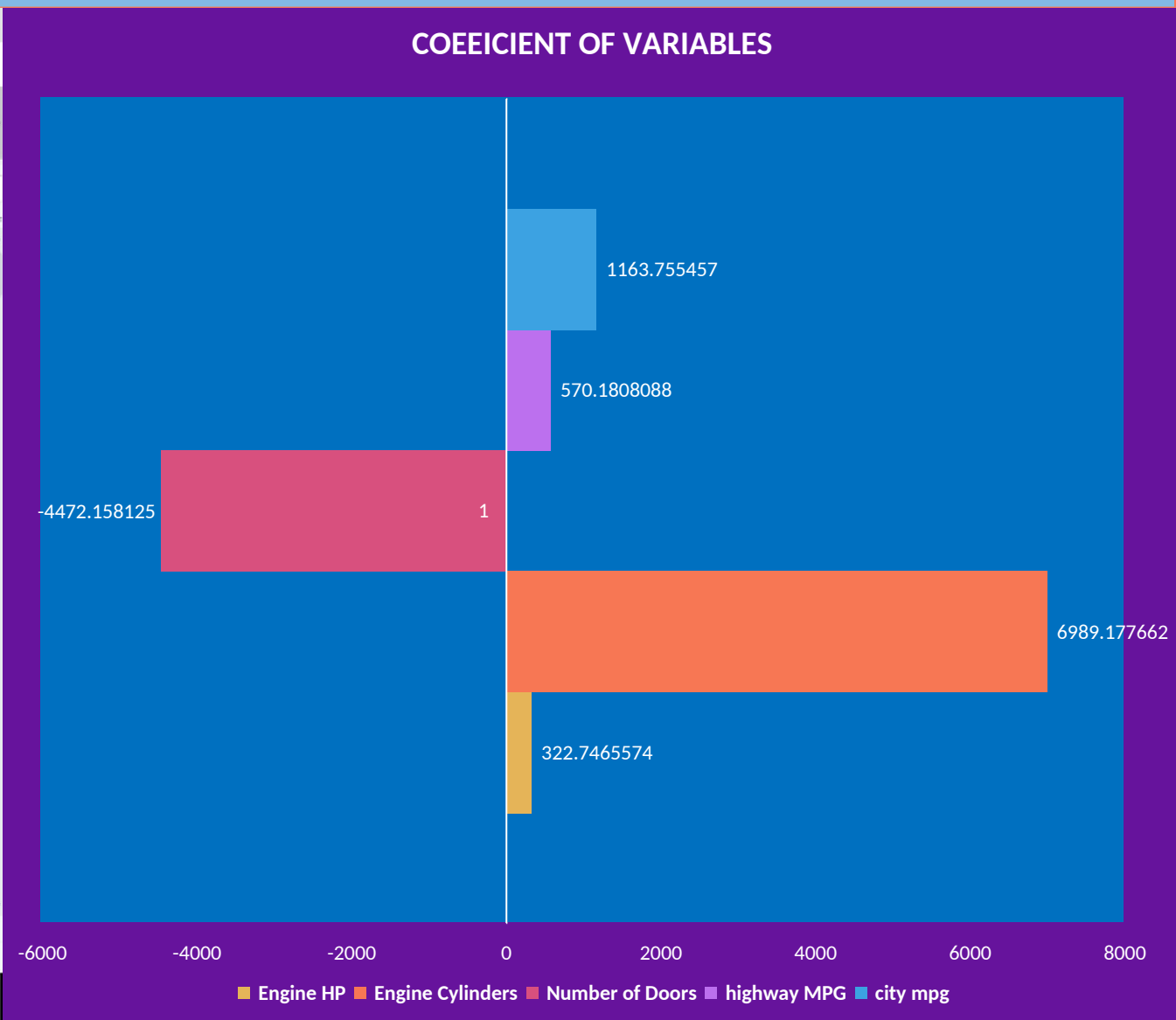
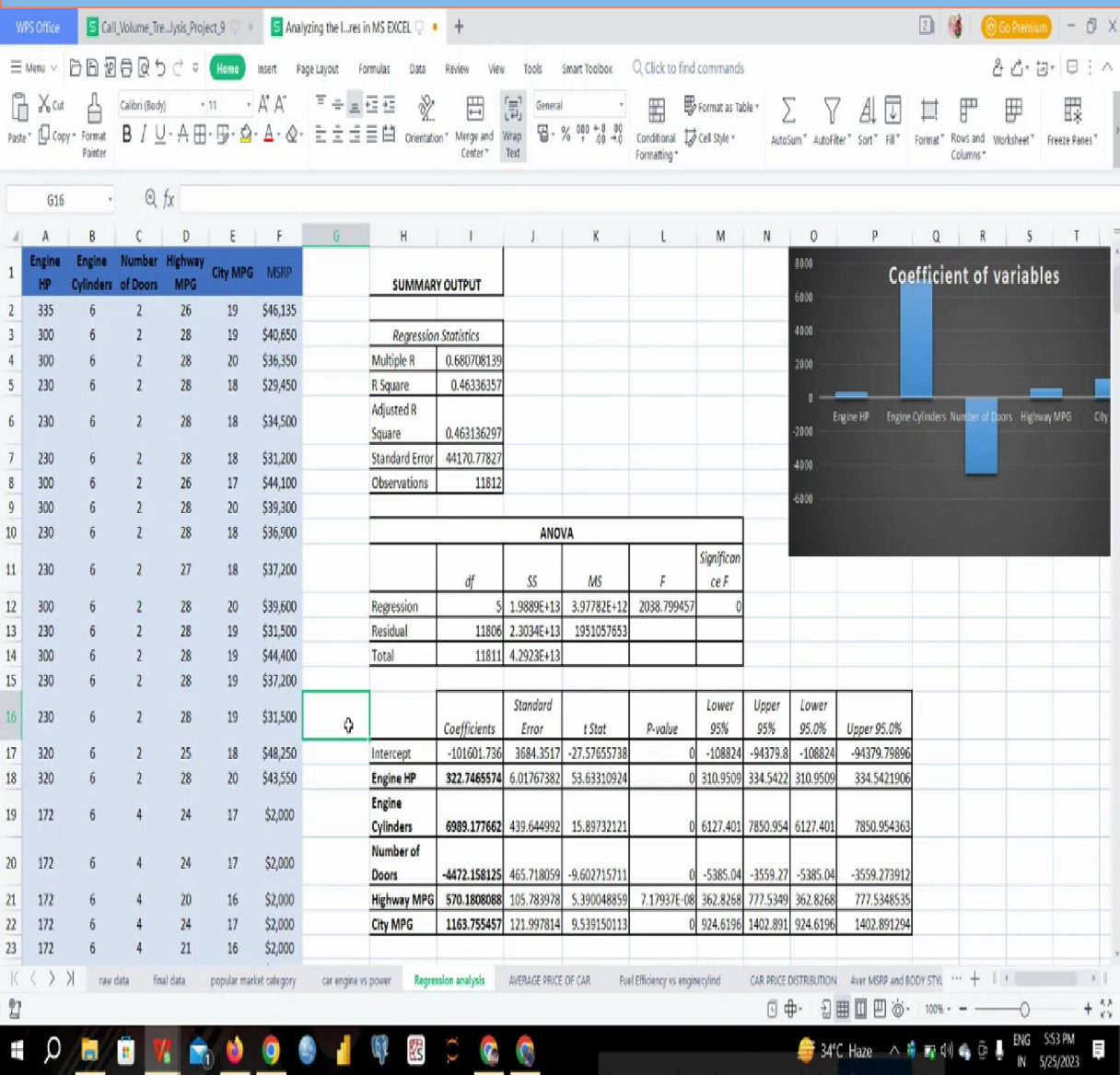
- Task 2: Create a scatter chart that plots engine power on the x-axis and price on the y-axis. Add a trendline to the chart to visualize the relationship between these variables.

Engine HP	MSRP
335	46135
300	40650
300	36350
230	29450
230	34500
230	31200
300	44100
300	39300
230	36900
230	37200
300	39600
230	31500
300	44400
230	37200
230	31500
320	48250
320	43550
172	2000
172	2000
172	2000
172	2000
172	2000
172	2000
172	2000
172	2000
172	2000
172	2000



Insight Required: Which car features are most important in determining a car's price?

- **Task 3:** Use regression analysis to identify the variables that have the strongest relationship with a car's price. Then create a bar chart that shows the coefficient values for each variable to visualize their relative importance.

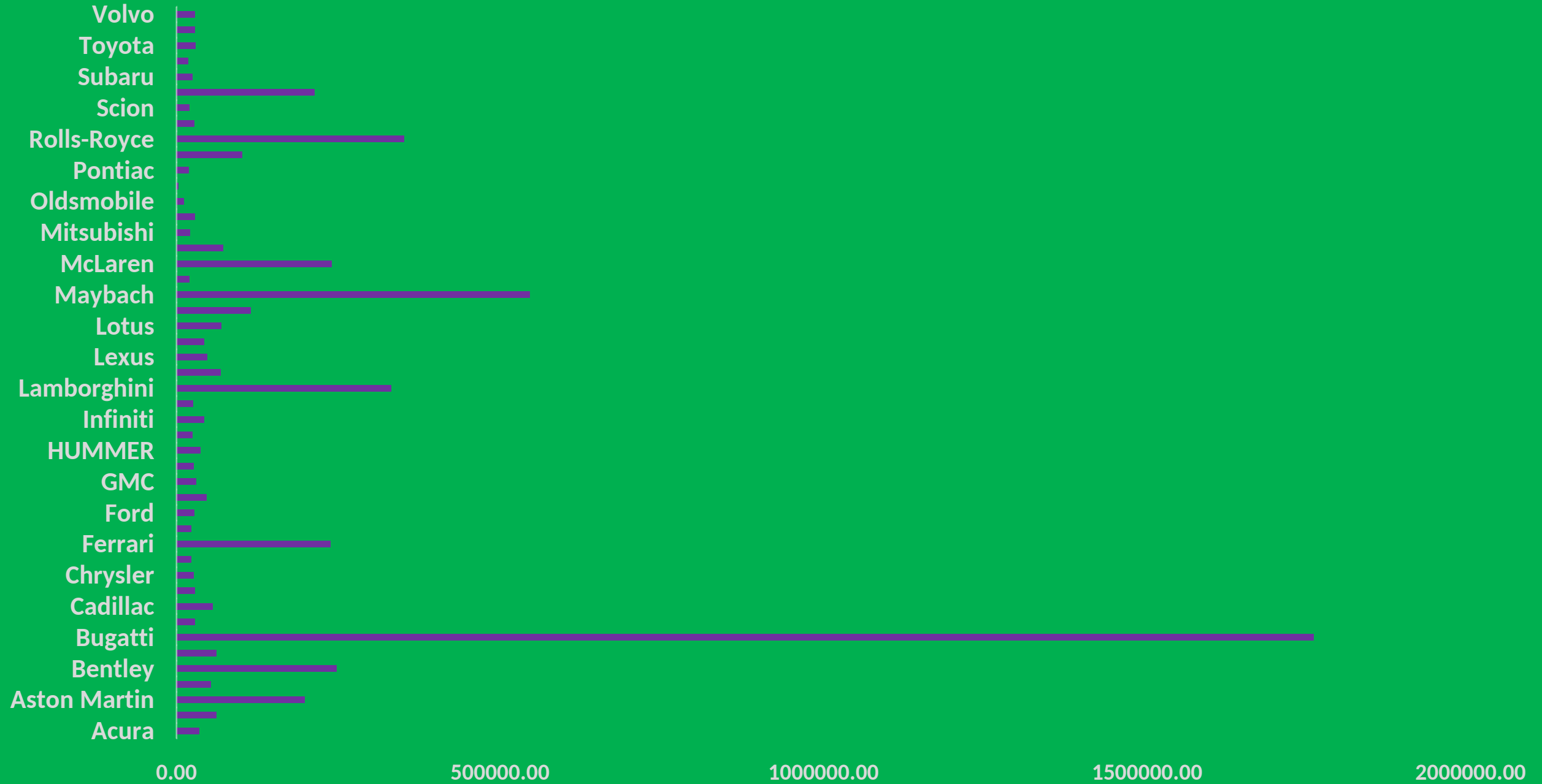


Insight Required: How does the average price of a car vary across different manufacturers?

- **Task 4.A:** Create a pivot table that shows the average price of cars for each manufacturer.
- **Task 4.B:** Create a bar chart or a horizontal stacked bar chart that visualizes the relationship between manufacturer and average price.

Row Labels	Average of MSRP	HUMMER	36464.41
Acura	34887.59	Hyundai	24597.04
Alfa Romeo	61600.00	Infiniti	42394.21
Aston Martin	197910.38	Kia	25112.39
Audi	53452.11	Lamborghini	331567.31
Bentley	247169.32	Land Rover	67823.22
BMW	61546.76	Lexus	47549.07
Bugatti	1757223.67	Lincoln	42494.37
Buick	28206.61	Lotus	69188.28
Cadillac	56231.32	Maserati	114207.71
Chevrolet	28273.36	Maybach	546221.88
Chrysler	26722.96	Mazda	19719.06
Dodge	22390.06	McLaren	239805.00
Ferrari	237383.82	Mercedes-Benz	71537.81
FIAT	22206.02	Mitsubishi	21215.47
Ford	27393.42	Nissan	28513.37
Genesis	46616.67	Oldsmobile	11542.54
GMC	30493.30	Plymouth	3122.90
Honda	26629.82	Pontiac	19321.55
		Porsche	101622.40
		Rolls-Royce	351130.65
		Saab	27413.50
		Scion	19932.50
		Spyker	213323.33
		Subaru	24827.50
		Suzuki	17900.96
		Toyota	28946.15
		Volkswagen	28076.20
		Volvo	28541.16
		Grand Total	40559.94

MANUFACTURERS VS AVG PRICE

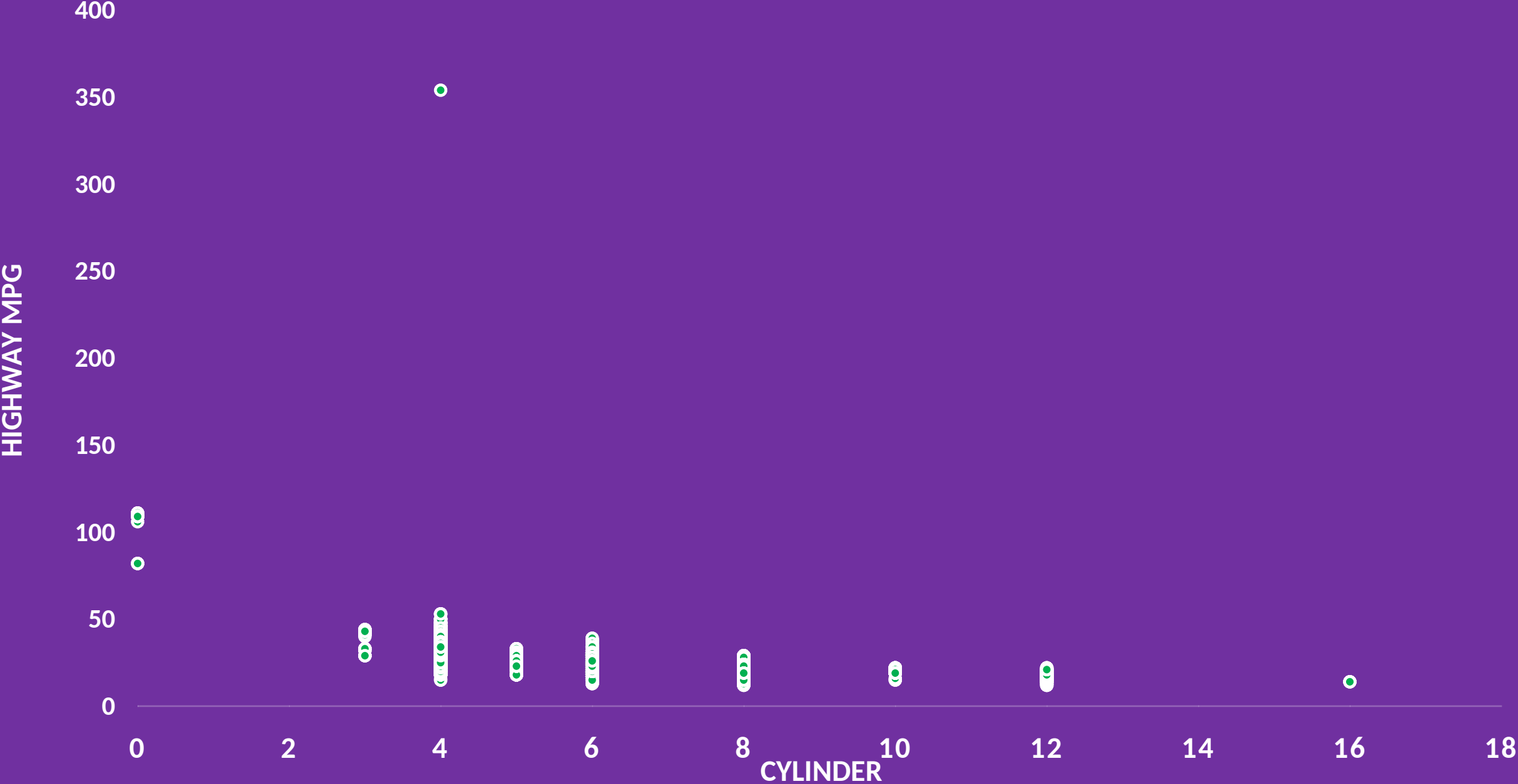


Insight Required: What is the relationship between fuel efficiency and the number of cylinders in a car's engine?

- **Task 5.A:** Create a scatter plot with the number of cylinders on the x-axis and highway MPG on the y-axis. Then create a trendline on the scatter plot to visually estimate the slope of the relationship and assess its significance.
- **Task 5.B:** Calculate the correlation coefficient between the number of cylinders and highway MPG to quantify the strength and direction of the relationship.

Engine Cylinders		highway MPG	
6		26	
6		28	
6		28	
6		28	
6		28	
6		28	
6		26	
6		28	
6		28	
6		27	
6		28	
6		28	
6		28	
6		28	
6		28	
6		25	
6		28	
6		24	
6		24	
6		20	
6		24	
6		21	
6		24	
CORRELATION COEFFICIENT		20	
-0.62	SHOWING NEGATIVE CORRELATION		

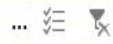
NO OF CYLINDERS VS HIGHWAY MPG



BUILDING DASHBOARD



PROJECT- ANALYSING THE IMPACT OF CAR FEATURES ON PRICE AND PROFITABILITY



Acura

Alf...

Ast...

Audi

Ben...

BMW

Bug...

Buick

Ca...

Che...

Chr...

Do...

Fer...

FIAT

Ford

Ge...

GMC

Ho...

HU...

Hy...

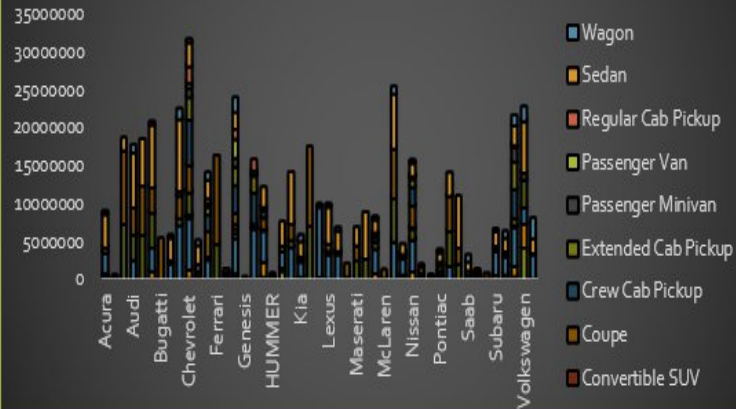
Infiniti

Kia

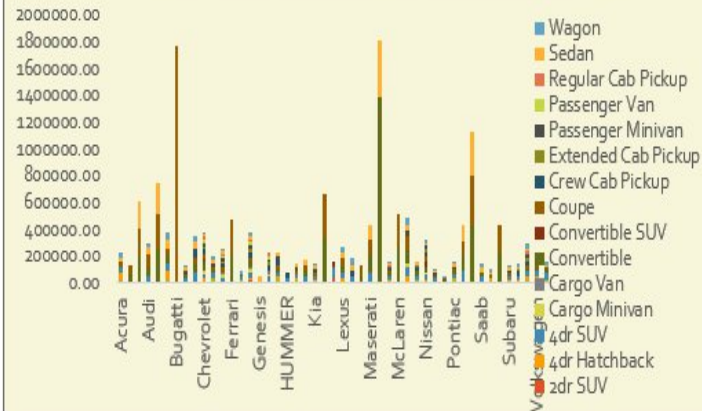
La...

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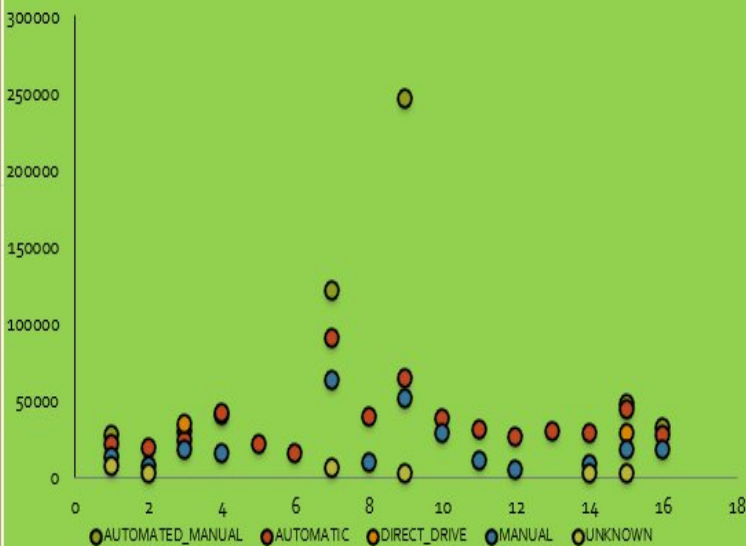
TOTAL CAR PRICE DISTRIBUTION



AVG PRICE, BRAND AND VEHICLE STYLE



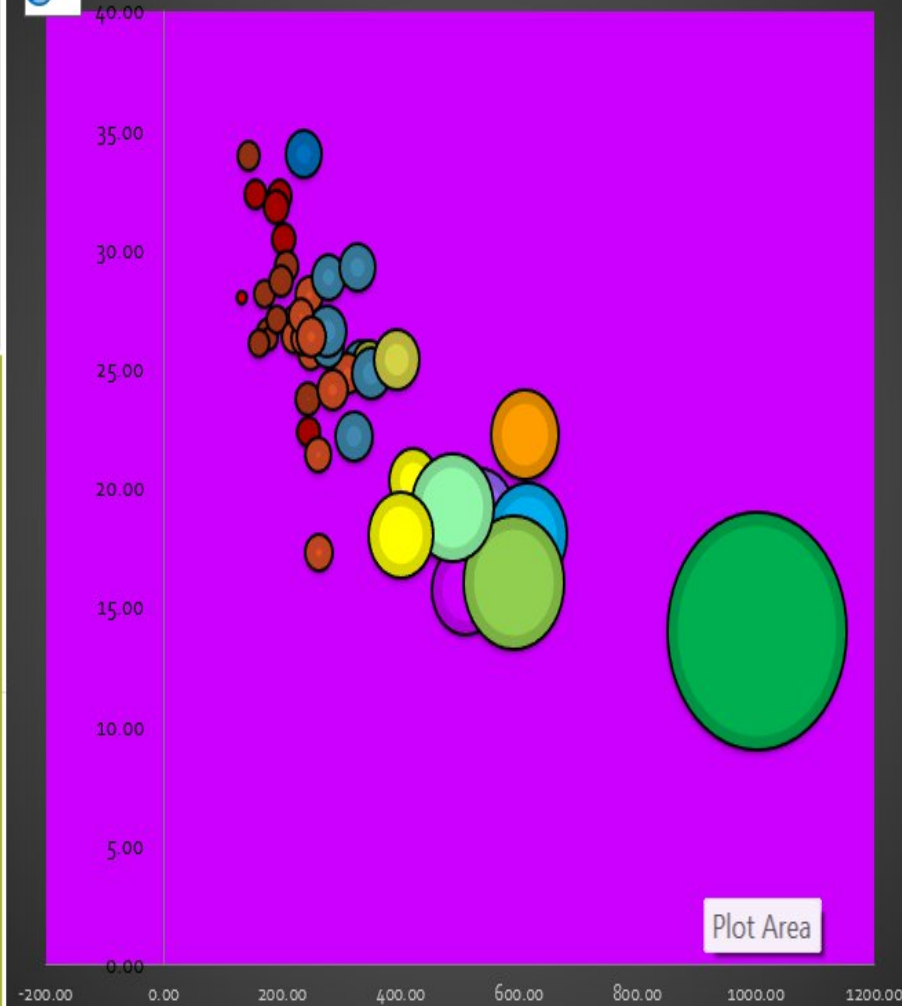
TRANSMISSION TYPE VS MSRP



YEAR VS AVG HIGHWAY MPG



BRAND WITH AVG OF HIGHWAY MPG, ENGINE HP, MSRP



ABC CALL VOLUME TREND ANALYSIS

PROJECT 8

PROJECT DESCRIPTION:

- In this project, we will move into the world of Customer Experience (CX) analytics, specifically focusing on the inbound calling team of a company. A dataset that spans 23 days and includes various details such as the agent's name and ID, the queue time (how long a customer had to wait before connecting with an agent), the time of the call, the duration of the call, and the call status (whether it was abandoned, answered, or transferred) is provided to us. The goal is to attract, engage, and delight customers, turning them into loyal advocates for the business.

APPROACH:

- To perform the project a systematic approach was followed. A dataset having relevant information on agent's name and ID, the queue time, the time of the call, the duration of the call, and the call status was downloaded. Microsoft Excel 2021 was selected as the primary tool for data analysis due to its versatility and robust capabilities in handling tabular data. There are N/A in the dataset but as per the tasks those data columns are not required so they remain there in the dataset which will not affect our analysis.

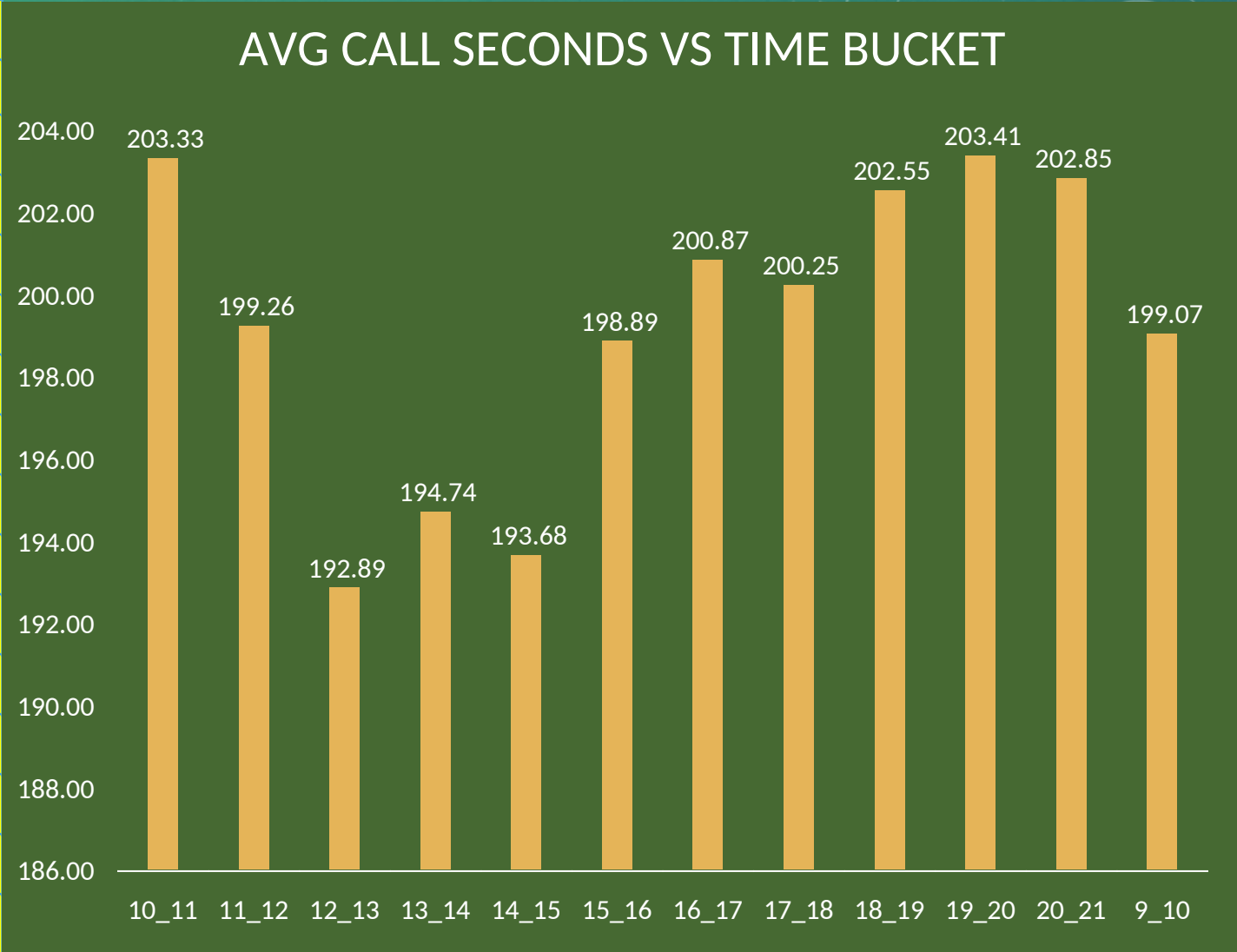
TECH-STACK USED:

- Software: Microsoft Excel 2021
- Purpose: Excel was chosen for its extensive data analysis functions, including pivot tables, charts, and statistical functions and visualization of the results.

AVERAGE CALL DURATION:

Your Task: What is the average duration of calls for each time bucket?

Call_Status	answered
Time Bucket	Average of Call_Seconds (s)
10_11	203.33
11_12	199.26
12_13	192.89
13_14	194.74
14_15	193.68
15_16	198.89
16_17	200.87
17_18	200.25
18_19	202.55
19_20	203.41
20_21	202.85
9_10	199.07
Grand Total	198.62



- As seen in the graph we can clearly the average duration of calls for each time buckets for answered calls only. For **7pm-8pm** the duration of calls is the **highest** and for **12 pm -1 pm** the value is the **lowest** amongst all

CALL VOLUME ANALYSIS:

Your Task: Can you create a chart or graph Highest no of calls received during between **11 am to 12 pm** as clearly seen in this

Time Bucket	Count of Customer Phone No	Count of Call Seconds (s)
10_11	13313	11.28%
11_12	14626	12.40%
12_13	12652	10.72%
13_14	11561	9.80%
14_15	10561	8.95%
15_16	9159	7.76%
16_17	8788	7.45%
17_18	8534	7.23%
18_19	7238	6.13%
19_20	6463	5.48%
20_21	5505	4.67%
9_10	9588	8.13%
Grand Total	117988	100.00%

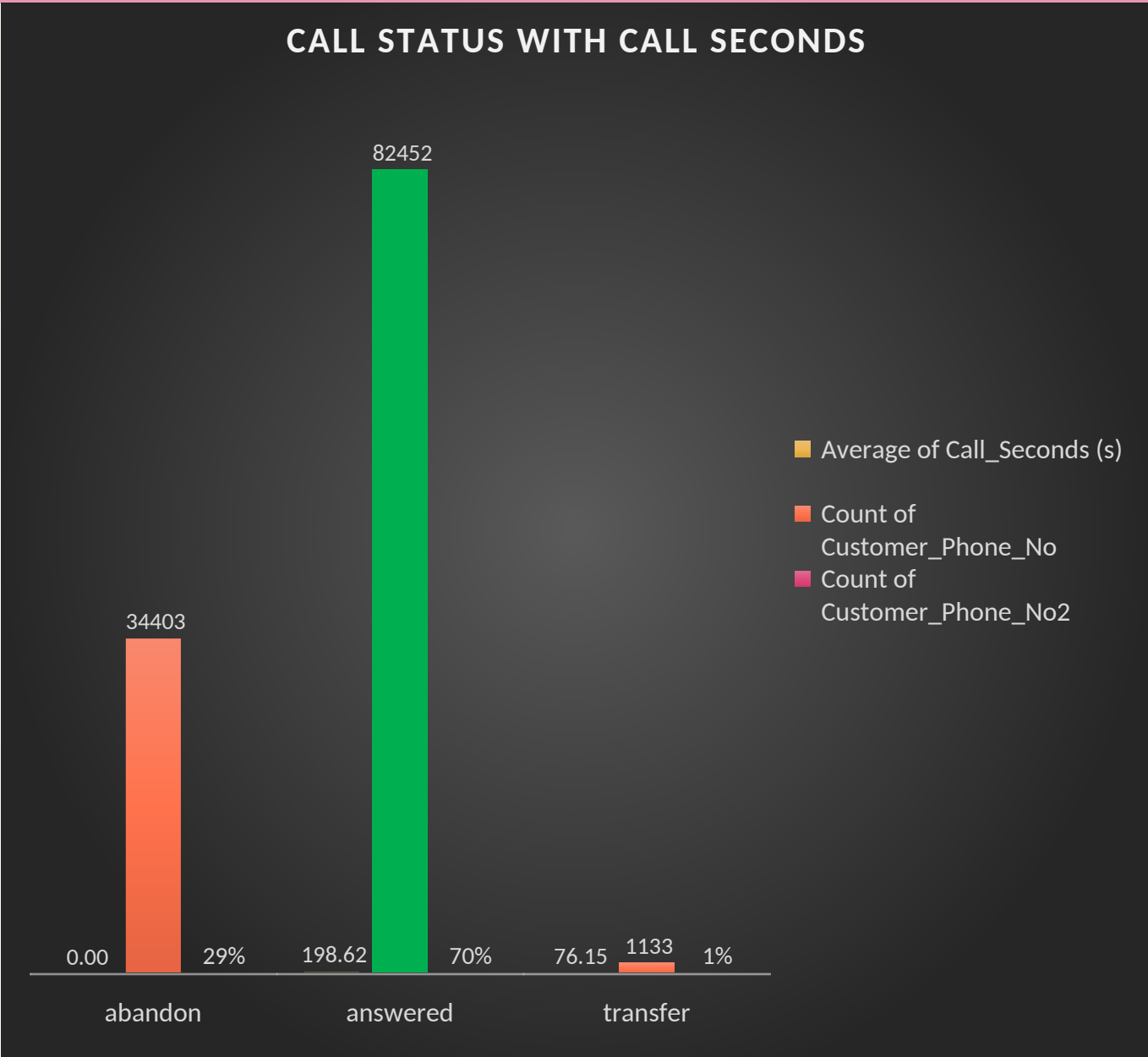


- Highest no of calls received during between **11 am to 12 pm** as clearly seen in this graph

MANPOWER PLANNING:

Your Task: What is the minimum number of agents required in each time bucket to reduce the abandon rate to 10%?

Row Labels	Average of Call_Seconds (s)	Count of Customer_Phone No	Count of Customer_Phone No2
abandon	0.00	34403	29%
answered	198.62	82452	70%
transfer	76.15	1133	1%
Grand Total	139.53	117988	100.00%



The current rate of abandoned calls is approximately 30%.

Propose a plan for manpower allocation during each time bucket (from 9 am to 9 pm) to reduce the abandon rate to 10%.

In other words, you need to calculate the minimum number of agents required in each time bucket to ensure that at least 90 out of 100 calls are answered.

Time Bucket	Count of Call_Seconds (s)	Count of Call_Seconds (s)2	call_seconds	Agent Required	
10_11	11.28%	0.11	0.11	6	
11_12	12.40%	0.12	0.12	7	
12_13	10.72%	0.11	0.11	6	
13_14	9.80%	0.10	0.10	6	
14_15	8.95%	0.09	0.09	5	
15_16	7.76%	0.08	0.08	4	
16_17	7.45%	0.07	0.07	4	
17_18	7.23%	0.07	0.07	4	
18_19	6.13%	0.06	0.06	3	
19_20	5.48%	0.05	0.05	3	
20_21	4.67%	0.05	0.05	3	
9_10	8.13%	0.08	0.08	5	
Grand Total	100.00%	1.00	1.00	57	

Date_&_Time2	01-Jan			
Row Labels				
	Sum of Call_Seconds (s)			
09	35313.00		188.0	Per one day
10	53087.00		38	As mentioned, 60%
11	67751.00	So	60%	38
12	72680.00		90%	57
13	59693.00			Total agent req
14	76137.00			
15	65689.00			
16	59464.00			
17	68155.00			
18	53096.00			
19	40141.00			
20	25281.00			
21	177.00			
Grand Total	676664.00			

- Total agent required are **57** and the **minimum number of agents** required in each time bucket to ensure that at least 90 out of 100 calls are answered are also shown by the table

NIGHT SHIFT MANPOWER PLANNING:

Your Task: Propose a manpower plan for each time bucket throughout the day, keeping the maximum abandon rate at 10%.

Count of Call Status	Column Labels				
Row Labels	abandon	answered	transfer	Grand Total	
01-Jan	684	3883	77	4644	
02-Jan	356	2935	60	3351	
03-Jan	599	4079	111	4789	
04-Jan	595	4404	114	5113	
05-Jan	536	4140	114	4790	
06-Jan	991	3875	85	4951	
07-Jan	1319	3587	42	4948	
08-Jan	1103	3519	50	4672	
09-Jan	962	2628	62	3652	
10-Jan	1212	3699	72	4983	
11-Jan	856	3695	86	4637	
12-Jan	1299	3297	47	4643	
13-Jan	738	3326	59	4123	
14-Jan	291	2832	32	3155	
15-Jan	304	2730	24	3058	
16-Jan	1191	3910	41	5142	
17-Jan	16636	5706	5	22347	
18-Jan	1738	4024	12	5774	
19-Jan	974	3717	12	4703	
20-Jan	833	3485	4	4322	
21-Jan	566	3104	5	3675	
22-Jan	239	3045	7	3291	
23-Jan	381	2832	12	3225	
Grand Total	24400	22450	4400	47250	
5130	TOTAL AVG CALLS				
1539	FOR NIGHT				
76	ADDITIONAL hr. REQ				
15	ADDITIONAL AGENT REQ				

ASSUMPTION: we have an assumption value as provided for call between 9pm-9am and their call distribution value				
CALL BETWEEN 9PM-9AM	CALL DISTRIBUTION	TIME DISTRIBUTION	AGENT REQUIRED	ROUND FIGURE
9PM-10PM	3	10	1.5	2
10PM-11PM	3	10	1.5	2
11PM-12PM	2	15	1	1
12PM-1AM	2	15	1	1
1AM-2AM	1	30	0.5	1
2AM-3AM	1	30	0.5	1
3AM-4AM	1	30	0.5	1
4AM-5AM	1	30	0.5	1
5AM-6AM	3	10	1.5	2
6AM-7AM	4	7.5	2	2
7AM-8AM	4	7.5	2	2
8AM-9AM	5	6	2.5	3
TOTAL	30		15	19

- Total **Average calls** are **5130**
- For night it was told that **30% of calls** so **30%*5130= 1539**
- Then for additional hour we calculate by round formula
=round (1539*198.6*0.9/3600,0), it will result into **76 hr**
- Then for agent who work **5 hr** in a day the we will, calculate by round formula **=round (76/5,0)** and we get **15 as additional agent** required for the night hour Planning.

RESULT: We have achieved so many fruitful insights about the ABC Call Volume Trend Analysis by doing so many tasks such as call status, call duration, man power planning with the help of Microsoft Excel and its useful functions which will help us get better idea about how a call volume analysis have been doing

LEARNINGS

- **Data Cleaning & Preparation:** Developed expertise in handling missing data, removing duplicates, detecting outliers, and ensuring data integrity for accurate analysis.
- **Exploratory Data Analysis (EDA):** Strengthened skills in summarizing datasets, performing univariate and bivariate analysis, and visualizing trends using statistical techniques.
- **SQL Querying & Optimization:** Gained proficiency in writing complex SQL queries, including joins, subqueries, CTEs, and window functions, to extract meaningful insights efficiently.
- **KPI & Metric Investigation:** Improved ability to analyze key business metrics, identify anomalies, and provide actionable recommendations to optimize business operations.
- **Pattern Recognition & Trend Analysis:** Developed the ability to recognize patterns in customer behavior, operational data, and financial trends to support strategic decision-making.
- **Data Visualization & Reporting:** Strengthened the ability to create meaningful dashboards and reports in Microsoft Excel, improving data-driven decision-making processes.
- **Industry-Specific Insights:** Gained domain knowledge in various industries such as social media analytics, hiring trends, banking, automotive, and operational efficiency.
- **Business Problem-Solving:** Enhanced problem-solving skills by identifying challenges in datasets, formulating hypotheses, and providing data-backed solutions.
- **Stakeholder Communication:** Improved ability to present data-driven findings in a clear and structured manner, ensuring stakeholders understand the insights and recommended actions.

THANK

YOU